

Evaluating the Impact of Discovered Causal Structures on Structure-Aware Shapley Methods: An Experimental Pipeline

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1 Introduction

Explainable AI has moved from a research preference to a deployment requirement, particularly across critical decisioning socioeconomic domains such as credit scoring, healthcare access, and public resource allocation, where decisions derived from a machine learning model must be justified to affected individuals and regulatory bodies [1].

While Shapley values derived from cooperative game theory have emerged as a gold standard for local feature attribution due to their unique axiomatic foundations, traditional implementations frequently rely on the simplifying assumption of feature independence. This assumption treats all feature permutations as equally probable during coalition formation, which can systematically misattribute credit when the true generative process involves structured causal dependencies among inputs [5, 7].

In scenarios that involve causal dependencies, recent literature has introduced structure-aware Shapley frameworks that explicitly embed topological and causal constraints into the attribution process. Asymmetric Shapley Values (ASV) restrict permutations to those consistent with a causal ordering [3]; Causal Shapley Values (CSV) replace observational marginalization with interventional conditioning based on Pearl’s do-calculus [5]; and Shapley Flow reframes credit assignment from nodes to directed edges in a causal graph [17]. Each of these methods promises attributions that are more faithful to the underlying causal mechanism that generated the data.

Historically, these structure-aware Shapley frameworks have been demonstrated almost exclusively on well-characterized systems, domains backed by years of research where the causal links between variables are firmly established and treated as ground truth. A different situation arises far more often in practice: a practitioner suspects that causal relationships exist among the features of a system but cannot point to a validated graph describing them. In this setting, a traditional Shapley computation will ignore those dependencies entirely, distributing credit as if every feature were an independent, interchangeable coalition partner. The practitioner is therefore caught between a method that disregards causal structure and a family of methods that assume it is already known.

The majority of structure-aware Shapley papers treat the causal graph as given or completely specified by a domain expert. In practical applications, the true causal graph is almost never directly observable. Practitioners must instead rely on automated causal discovery algorithms such as the constraint-based PC algorithm [15] and the functional causal model DirectLiNGAM [14] to infer a plausible graph structure from observational data. These estimated graphs are inherently noisy, subject to false-positive and false-negative edge errors, and sensitive to violations of their underlying assumptions [4]. Different discovery algorithms applied to the same data can produce

structurally divergent graphs, and this work directly measures how much that algorithmic disagreement propagates into the final feature attributions, even when neither graph is correct.

The gap this work fills is not simply a missing application study but the absence of a measurement methodology. No existing framework in the structure-aware Shapley literature quantifies how attribution quality degrades as a function of graph imperfection, and without such a framework a practitioner has no principled basis for deciding whether a discovered graph improves or distorts explanations relative to the simpler graph-free baseline [2]. The evaluation design introduced here is built to be general: neither Magnitude Divergence nor Sign Disagreement presupposes anything about the data distribution, the predictive model, or the discovery algorithm used, so both metrics apply directly to any setting where a structure-aware Shapley method and an imperfect graph are combined. This generality matters because the underlying problem is universal. Any practitioner who suspects causal relationships among their features and considers using a structure-aware Shapley method faces the same open question: does graph injection actually improve the explanation, and is the improvement robust enough to be trusted in a consequential decision? Without a replicable measurement framework, that question has no systematic answer. The metrics defined here give it one, and the experimental results that follow demonstrate their diagnostic value across two datasets with structurally different discovery failure modes.

To articulate that framework, we built an experimental pipeline that feeds automatically discovered graphs into three structure-aware Shapley methods and measures, for the first time, how much attribution quality degrades when the graph is imperfect. The pipeline evaluates three structure-aware Shapley methods (Asymmetric Shapley, Causal Shapley, and Shapley Flow) under three causal graph sources: the PC-discovered graph, the LiNGAM-discovered graph, and the True DAG used as an oracle reference. Evaluation is performed on two benchmarks: a controlled synthetic dataset with 50 features and known linear causal structure, and the real-world Sachs cell signaling dataset [11], where an established consensus DAG allows external validation.

The central research question is: when a practitioner feeds an imperfect discovered causal graph into a structure-aware Shapley method, does the method produce attributions that meaningfully differ from a graph-free baseline, and do those attributions move closer to or further from the oracle attributions obtained with the True DAG? The answer has direct implications for how causal explainability tools should be adopted in practice.

To answer this question consistently, the analysis tracks two dimensions along which an attribution can shift when causal structure is injected: the *magnitude* of the attribution assigned to a feature and the *direction* (sign) of that attribution. This separation

is deliberate. Prior comparisons of attribution methods overwhelmingly report how much the size of the attributions changes, while the question of whether a feature’s contribution flips from pushing a prediction up to pushing it down is reported far less often, even though a sign reversal is arguably the more consequential failure for a practitioner trying to explain a single decision. A graph that merely rescales attributions still preserves the qualitative story told to an affected individual; a graph that inverts signs tells the opposite story. Both dimensions therefore matter, and they can move independently: a method may keep magnitudes stable while reordering signs, or preserve signs while inflating magnitudes. A magnitude change alone probably never crosses the sign boundary: it can only amplify or suppress the size of an attribution while keeping it on the same side of zero; a sign flip is the distinct event of an attribution crossing zero, which can happen even when the accompanying magnitude shift is small. The two dimensions are operationalized by a single metric each, used identically across every comparison: a Magnitude Divergence and a Sign Disagreement, each reported relative to a named reference (the graph-free baseline, the oracle DAG, or the opposite discovered graph). The mathematical logic of both metrics are detailed deeply in Section 3.5. Thanks to the fact that this dual-axis approach operates independently of any specific dataset or model architecture, it establishes a universal, replicable framework for future work; for now, setting magnitude and direction as our primary analytical axes provides the definitive standard needed to assess the viability of any structure-aware method in real-world scenarios.

2 Conceptual Framework and Methodology

2.1 Model Explainability and Causal Understanding

Model explainability, as used throughout this thesis, refers to the goal of making transparent why a trained predictive model produced a particular output for a particular input, not to recovering the governing laws of the physical system the model was trained on. This distinction mirrors the conceptual separation established by Baron [1] between causal understanding of nature and causal understanding of a model’s decision function, and it determines what counts as a valid explanation: the question is not what the world did but what the model did, and why it did it.

In practical machine learning scenarios, conducting physical manipulations or randomized controlled treatments is often expensive, time-consuming, or ethically impossible. Consequently, model explanations must be grounded in the observational data distribution used during training, while ideally respecting the causal relationships that govern how features jointly determine model outputs. Structure-aware Shapley methods operationalize this grounding by encoding causal relationships as constraints on the

attribution computation, rather than by performing physical interventions.

Explainability methods differ in scope. Global methods describe average model behaviour across a population; local methods explain a single prediction for a single instance. This thesis focuses on local attribution: assigning each input feature a numerical credit for its contribution to why this particular prediction came out the way it did. Shapley values are the standard tool for this task. They are model-agnostic, work with any black-box predictive function, and satisfy the Efficiency axiom, meaning the credits always sum exactly to the gap between this instance’s prediction and the model’s average output [7].

The limitation of Traditional Shapley is the Symmetry axiom. It treats every feature as an interchangeable coalition partner by averaging over all possible orderings with equal weight. When features have causal dependencies in the real system, this is wrong: whenever x_i causally influences x_j , symmetric averaging inflates x_j ’s credit at the expense of its upstream driver x_i , because the correlation between the two gets distributed as if neither caused the other. The structure-aware Shapley methods described in the remainder of this chapter are designed to correct precisely this.

2.2 Counterfactual Explanations and the Structural Causal Model Framework

A natural complement to attribution-based explanations is the counterfactual paradigm, which approaches the same question from a different direction: rather than assigning each feature a credit value, it asks what minimal change to the inputs would have produced a different output, placing the explanation in the language of alternative scenarios. This section examines why counterfactual methods face causal feasibility constraints in practice, how the Structural Causal Model (SCM) framework addresses those constraints, and why the SCM also provides the theoretical underpinning for Shapley Flow.

Counterfactual explanations provide an intuitive contrastive explainability framework by identifying the minimal input feature perturbations required to alter an algorithmic output [16]. Standard optimization methods, however, generate counterfactuals that may violate physical plausibility constraints by placing instances in low-density regions of the joint feature space or proposing changes that contradict causal relationships [8].

To satisfy real-world feasibility constraints, explanations must be grounded within a Structural Causal Model (SCM), formally defined as a tuple $\mathcal{G} := (\mathcal{S}, P(\varepsilon))$, where $\mathcal{S}$ is a collection of deterministic structural equations of the form $x_i = f_i(\text{Pa}(x_i), \varepsilon_i)$, and $P(\varepsilon)$ is a joint distribution over independent exogenous noise variables [9]. Full abduction-action-prediction within an SCM is computationally demanding and requires

complete structural equation knowledge. Structure-aware Shapley methods circumvent this bottleneck by using the causal graph topology to define conditional distributions for feature imputation, approximating the post-interventional distribution without requiring full structural equation reconstruction.

The original Shapley Flow [17] requires approximating structural equations to propagate values along edges. This implementation replaces that with a binary activation rule (detailed in Section 2.5.3) so the method depends only on graph topology, not on a calibrated SCM. This approach preserves the framework’s core conceptual contribution, attributing credit to directed edges rather than isolated nodes, while ensuring tractable computation that relies solely on the discovered graph topology and trained predictive model.

2.3 Causal Discovery: Structure Identification

The structure-aware Shapley methods discussed in this chapter all require a causal graph as input, but in most real-world applications that graph is neither supplied by domain experts nor directly observable. Causal discovery algorithms fill this gap by inferring a plausible graph structure from observational data alone, without requiring controlled experiments or prior structural knowledge. This pipeline employs two of the most widely used and contrasting approaches, one constraint-based and one functional-model-based, so that the downstream experiments cover the range of graph-quality failures a practitioner is likely to encounter.

PC Algorithm [15]. A constraint-based method that recovers the causal skeleton through conditional independence tests (Fisher-z at $\alpha = 0.05$) and applies Meek’s orientation rules to produce a Completed Partially Directed Acyclic Graph. Undirected edges are resolved by orienting them from lower to higher feature index, yielding a valid DAG for downstream Shapley computation.

DirectLiNGAM [14]. A functional causal model that exploits linear non-Gaussian structures to simultaneously identify causal ordering and structural coefficients. Each entry in the estimated adjacency matrix represents the linear weight a parent contributes to a child. Because DirectLiNGAM initially returns a dense matrix containing minor estimation noise, a magnitude threshold is applied to recover a sparse structure. Edges with an absolute coefficient below 0.10 are pruned; this cutoff effectively removes estimation noise while retaining edges that carry a structurally meaningful direct effect.

Both algorithms operate strictly on the X -only training partition, excluding the target for the discovery step. While both the synthetic dataset and the real-world biological data likely do not perfectly fulfill all the theoretical requirements of these algorithms,

this imperfection is deliberate. This is a realistic robustness test, reflecting the natural unmeasured confounding and structural uncertainty practitioners face in real-world applications.

2.4 Traditional Shapley Values and the Axiomatic Framework

Before introducing the structure-aware extensions that modify this foundation, it is necessary to establish precisely what Traditional Shapley values compute and what their axiomatic guarantees mean in practice. Traditional Shapley is the method every practitioner already uses and the reference point against which every structure-aware method is benchmarked throughout this thesis; understanding exactly where the Symmetry axiom fails under causal structure is also what makes each extension’s design choice interpretable.

Traditional Shapley values originate in cooperative game theory as the unique solution to the problem of fairly dividing the total payoff of a game among the players who produced it [13]. In the explainability transfer of this idea, the players are the input features, the coalition is any subset S of features whose values are known, and the payoff of a coalition is the model output produced when only those features take their actual instance values while the remaining features are treated as absent. A feature that is in the coalition contributes its real value x_j ; a feature that is out of the coalition is marginalized over the background distribution, so the model is evaluated as if that feature’s value were unknown and drawn from the data at large. The marginal contribution of a feature is then the change in the payoff caused by moving that feature from outside the coalition to inside it, that is, the difference $f(S \cup \{i\}) - f(S)$ between the model output with the feature added and the model output without it.

Because this marginal contribution depends on which features are already present, the Shapley value averages it over every order in which the features could be added to the coalition. Concretely, for a given permutation of the features each feature is added one at a time, its marginal contribution against the coalition of all features preceding it in that order is recorded, and the Shapley value of a feature is the average of these marginal contributions taken over all $N!$ permutations. Averaging over all orderings is precisely what makes the allocation fair in the game-theoretic sense and is what guarantees the four axioms; the construction satisfies **Efficiency** (the attributions sum to the gap between the prediction for the instance and the average prediction, $f(x)$ minus the baseline), **Linearity** (attributions from model ensembles combine linearly), **Null Player** (zero attribution for features with no marginal contribution in any coalition), and **Symmetry** (two features that contribute identically to every coalition receive equal credit). The Symmetry axiom is too restrictive for causal systems: it treats all features as interchangeable coalition partners, weighting every ordering equally, which

misattributes credit from downstream effects to upstream causes when the two are correlated through the causal structure.

2.5 Structure-Aware Shapley Methods: Theoretical Foundations

Structure-aware Shapley frameworks preserve Efficiency, Linearity, and Nullity while modifying or relaxing Symmetry to accommodate causal topologies.

2.5.1 Asymmetric Shapley Values (ASV)

ASV breaks the Symmetry axiom by replacing the uniform distribution over feature orderings with a weighting scheme $w(\pi)$ that places probability mass exclusively on permutations consistent with a partial causal ordering [3]. Concretely, the causal DAG is used to derive a topological order over the features, and only those permutations that respect this order, in which every ancestor appears before each of its descendants, are admitted into the averaging. This restriction shrinks the permutation universe relative to Traditional Shapley: instead of averaging over all $N!$ orderings, ASV averages only over the valid linear extensions of the partial order encoded by the graph, a strict subset whose size shrinks as the graph becomes more connected. Ancestral causes therefore always precede their downstream effects during coalition formation, preventing descendant features from receiving credit for contributions mediated by their ancestors.

Apart from this restriction on the ordering, ASV is mechanically identical to Traditional Shapley. The notion of a coalition is unchanged, features inside the coalition take their instance values and features outside it are marginalized over the background distribution, and the marginal contribution is still the same difference $f(S \cup \{i\}) - f(S)$ evaluated through the model. ASV retains observational marginalization for absent features, constraining only the permutation space and not the imputation distribution; the DAG enters the computation solely through the set of admissible orderings.

Formally, the attribution is the expectation of a feature’s marginal contribution taken over orderings drawn uniformly from the topological orderings of the feature DAG:

$$\phi_i(\text{ASV}) = \mathbb{E}_{\pi \sim U(\Pi_{\text{topo}})} [v(P_i^\pi \cup \{i\}) - v(P_i^\pi)], \quad (1)$$

where Π_{topo} is the set of all topological orderings (linear extensions) of the feature DAG, U is the uniform distribution over that set, P_i^π is the set of features preceding i in ordering π , and the value function $v(\cdot)$ is exactly the observational coalition value of Traditional Shapley, $v(S) = \mathbb{E}[f(X) \mid X_S = x_S]$, estimated by marginalizing the absent features over the background data. The only change relative to Traditional Shapley

is the replacement of the universe of all $N!$ orderings by the subset Π_{topo} consistent with the DAG. Because every feature still appears in every admissible ordering, no feature is forced to a structural-zero attribution. In our implementation, a uniform linear extension is drawn by a randomized Kahn procedure that maintains a pool of “ready” nodes whose ancestors have all been placed and selects one uniformly at each step, which respects every ancestor-descendant pair simultaneously and costs $O(N)$ per ordering.

2.5.2 Causal Shapley Values (CSV)

CSV retains a symmetric formulation over permutations but redefines the value function using interventional distributions rather than observational conditioning [5]. The coalition and averaging machinery is the same as in Traditional Shapley; what changes is how the absent features are filled in. Where Traditional Shapley conditions on the coalition features, preserving the correlations a confounder induces, Causal Shapley intervenes on them with Pearl’s do-operator:

$$v(S) = \mathbb{E}[f(X) \mid X_S = x_S] \quad (\text{Traditional}) \quad \text{vs} \quad v_{\text{do}}(S) = \mathbb{E}[f(X) \mid \text{do}(X_S = x_S)] \quad (\text{Causal}). \quad (2)$$

Intervening cuts the incoming edges of the features in S , so a feature is credited only for effects that flow through its own outgoing causal paths and not for indirect effects inherited from upstream correlations. The node-level attribution keeps the standard Shapley weighting:

$$\phi_i(\text{CSV}) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v_{\text{do}}(S \cup \{i\}) - v_{\text{do}}(S)]. \quad (3)$$

To make the interventional value computable, the theoretical formulation of CSV supports explicitly modeling shared hidden causes by merging bidirected confounder pairs into joint components. However, to ensure computational reproducibility across all experiments, explicit confounder information is not supplied for either the synthetic or the Sachs dataset in this implementation. Instead, every feature is treated as an independent component. This approach streamlines the architecture while strictly preserving the core do-intervention sampling across the network.

The computation relies on two distinct levels of stochasticity. An outer loop draws, for each Shapley trial, a fresh uniform random linear extension of the full feature DAG. An inner loop then estimates $v_{\text{do}}(S)$ from M Monte Carlo draws of the post-interventional distribution $P(X \mid \text{do}(X_S = x_S))$. The inner loop always traverses the features in a fixed deterministic topological order, because propagating an intervention correctly requires each node’s parents to be resolved before the node itself.

Each post-interventional draw fixes the intervened features to their instance values and then, proceeding in the fixed topological order, fills in the remaining features from their respective parents. Because confounder components are disabled, the missing features are drawn sequentially using the closed-form univariate Gaussian expression under a multivariate-Gaussian approximation $X \sim \mathcal{N}(\mu, \Sigma)$:

$$\mu_{A|B} = \mu_A + \Sigma_{AB}\Sigma_{BB}^{-1}(x_B - \mu_B), \quad \Sigma_{A|B} = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}. \quad (4)$$

By executing this continuous interventional sampling natively across the DAG, the method effectively severs non-causal information flow without requiring the immense computational overhead of multivariate confounded-component tracking.

2.5.3 Shapley Flow

Shapley Flow moves from node-based to edge-based attribution [17]. It extends the classical Shapley axioms to directed graph edges and introduces a Boundary axiom: the sum of attribution flow entering any intermediate node equals the sum leaving it, ensuring credit conservation at intermediate nodes. The system is viewed as a connected graph running from source features through intermediate features to the target Y , and the game asks how much each edge contributes to moving the prediction from its background value to its foreground value. The players are therefore the edges, not the features, and each edge receives the average marginal contribution it makes as edges are activated in random order:

$$\phi_{u \rightarrow v} = \mathbb{E}_{\pi \sim U(\Pi_E)} [V(E_{u \rightarrow v}^\pi \cup \{u \rightarrow v\}) - V(E_{u \rightarrow v}^\pi)], \quad \phi_u = \sum_{v: u \rightarrow v \in E} \phi_{u \rightarrow v}, \quad (5)$$

where Π_E is the set of orderings of the edge set E , $E_{u \rightarrow v}^\pi$ is the set of edges preceding $u \rightarrow v$ in ordering π , $V(\cdot)$ is the system value of a set of active edges, and node-level importance is recovered by summing each node’s outgoing edge credits. The system value assigns every node either its foreground value, taken from the instance, or its background value: a node takes its foreground value if it is a source with at least one active outgoing edge, or if it has at least one active incoming edge, or if its direct edge to Y is active; otherwise it takes its background value, and Y itself is always stripped before the model is evaluated. With this rule, the empty edge set yields the background prediction and the full edge set yields the foreground prediction, so the edge credits satisfy efficiency exactly: the sum of all edge credits equals $f(x_{\text{fg}}) - f(x_{\text{bg}})$. Because credit is routed along edges, graph errors propagate along all paths connected to an affected edge, not only at the node level.

The DAG plays a more central role here than in either ASV or CSV, because its edges are the players: with no graph there is no game to play. In the faithful formulation

of Wang et al. [17], the graph additionally supplies the structural mechanism between connected nodes, propagating each parent’s value through the corresponding causal function $f_{\text{parent} \rightarrow \text{child}}(x_{\text{parent}})$ when a node lies outside the active edge set; this couples a node’s attribution to its entire upstream sub-graph and is the source of the method’s computational cost. The implementation in this thesis preserves the edge-as-player formulation and the efficiency property but, as discussed in Section 2, replaces the full structural-equation propagation with the lighter binary foreground/background activation rule above, so that the method depends only on the discovered graph and the trained model. A final implementation detail involves how edges are ordered during the sampling process. Instead of forcing edges to activate in a strict causal sequence (from upstream roots to downstream leaves), they are shuffled entirely at random. Randomly shuffling the edges ensures that, roughly half the time, an intermediate feature’s effect on the target is evaluated before its parents update it. This allows the framework to fairly capture the full direct impact of intermediate features, preventing root causes from absorbing all the credit.

3 Experimental Setup and Practical Implementation

The proposed experimental pipeline to help practitioners navigate scenarios where measured systems contain underlying causal connections or highly correlated features is outlined below. Within this framework, two distinct datasets are evaluated across multiple structure-aware Shapley methods. The resulting variations in feature attributions are measured along two primary dimensions: magnitude deviation and sign disagreement, both of which are critical for ensuring the reliability of local explanations.

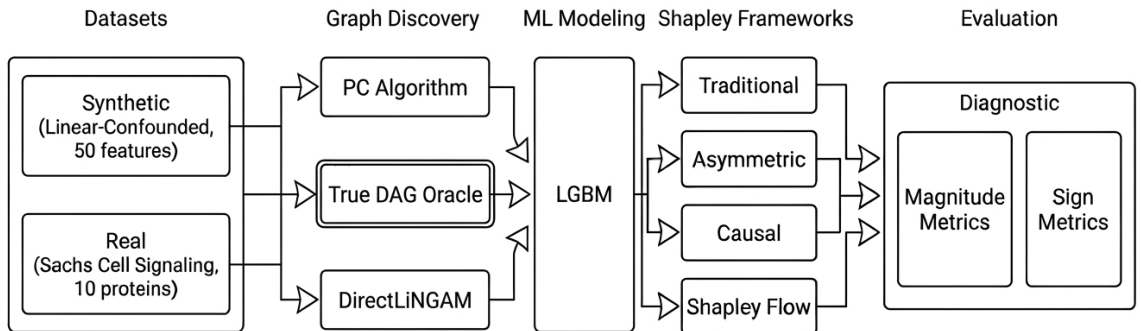


Figure 1: Overview of the experimental pipeline. End-to-end flow from data generation and causal discovery through Shapley computation to evaluation, illustrating how the two datasets, three graph sources, and three structure-aware methods feed into the magnitude and sign metrics.

3.1 Data Synthesis and Target Datasets

The experimental pipeline employs a synthetic benchmarking framework to generate a dataset with a known, deterministic ground-truth causal structure. Specifically, the analysis in this thesis relies on a simulated linear system with confounding, paired alongside the real-world Sachs dataset. This linear configuration was selected because it closely matches the operating assumptions of the two discovery algorithms: both the PC algorithm and DirectLiNGAM are designed around linear structural relationships. By utilizing a linear data-generating process, the discovery task remains aligned with what these estimators can in principle recover, ensuring that the errors observed downstream can be attributed to confounding and finite-sample noise rather than a fundamental mismatch between the data and the estimators’ functional assumptions. The deliberate addition of hidden confounders then violates the causal sufficiency assumption shared by both algorithms, providing precisely the stress condition of interest. The two core cases analyzed throughout the pipeline are therefore the following:

Linear System with Confounding (Synthetic). Strictly linear structural equations $X_j = \sum_{i \in \text{Pa}(j)} w_{ij} X_i + \varepsilon_j$, with $\varepsilon_j \sim \mathcal{N}(0, 0.5)$ and coefficients drawn from $\text{Uniform}(0.5, 2.0)$ with random sign. The graph is Erdős–Rényi with edge probability $p = 0.07$, yielding 87 $X \rightarrow X$ edges across 50 features (≈ 1.74 edges per node). Exactly 15 of 50 features are direct causal parents of Y (`y_parents_ratio` = 0.30), with the $X \rightarrow Y$ relationships following the same linear form as the $X \rightarrow X$ edges. Five hidden confounders each additively influence 2–3 features, violating causal sufficiency for both discovery algorithms. Dataset parameters: $N_{\text{FEATURES}} = 50$, $N_{\text{SAMPLES}} = 1000$, `random_state` = 42.

Sachs Cell Signaling Dataset (Real Data). 7,466 simultaneous measurements of 11 protein concentrations in stimulated human T-cells [11], obtained from the Carnegie Mellon University Philosophy Department causal datasets repository [12]. Akt kinase (`akt`) is the regression target Y and 10 proteins are input features (`raf`, `mek`, `plc`, `pip2`, `pip3`, `erk`, `pka`, `pkc`, `p38`, `jnk`). A consensus reference DAG has 17 $X \rightarrow X$ edges and 3 direct $X \rightarrow Y$ edges (`pip3->akt`, `pka->akt`, `erk->akt`). Raw concentrations are preprocessed in two steps, first $\log_{1p}$ transformation is applied to the entire dataset before the train/test split and secondly, after splitting, a Standard Scaler is fit exclusively on the training partition and then applied to both train and test sets, bringing all proteins to $\mu=0$, $\sigma=1$; this ensures LiNGAM’s $|\text{coef}| < 0.10$ pruning threshold is scale-consistent across all proteins regardless of raw concentration range, and prevents any information from the test set from leaking into the scaling statistics. Both datasets use an 80/20 train/test split with random seed 42. The synthetic graph

density ($\sim 7.1\%$, 1.74 edges/node) was calibrated to match the Sachs network density, enabling direct comparison between the two experimental tracks.

3.2 Predictive Model

A LightGBM gradient-boosted tree regressor is fitted to each dataset using Optuna-based hyperparameter optimization (50 TPE trials, RMSE objective). Feature selection is disabled for both datasets so that all input features participate in the model, keeping the feature space consistent with the causal graph and Shapley computation.

Table 1: Predictive Model Performance and Output Distribution on Test Sets. The output std $\hat{\sigma}$ is the standard deviation of the LightGBM predictions, used to normalise the magnitude metrics (Section 3.5). AU = Akt concentration units.

Dataset	R^2	RMSE	MAE	Output mean	Output std $\hat{\sigma}$
Linear-Conf (synthetic)	0.900	1.72	1.36	≈ -0.47	4.71
Sachs (real data)	0.679	88.78	17.82	≈ 81.2 AU	116.76 AU

Beyond accuracy, $\hat{\sigma}$ is recorded as a first-class model summary because all magnitude metrics in Section 3.5 are reported as a fraction of it. The two datasets live on incomparable raw scales ($\hat{\sigma} = 4.71$ for the synthetic target vs. $\hat{\sigma} = 116.76$ Akt units for Sachs); dividing by $\hat{\sigma}$ makes a Magnitude Divergence of 0.10 mean the same thing on both tracks, a divergence of 10% of the model’s explainable output spread. The formal justification for this choice of normaliser is given in Section 3.5.

3.3 Causal Discovery Framework

3.3.1 DirectLiNGAM Integration

The pipeline uses the `causal-learn` package [18]’s DirectLiNGAM implementation. The library outputs an adjacency matrix where `matrix[i,j]` is the coefficient of X_j on X_i ; the matrix is transposed before binarization. Edges with $|\text{coefficient}| < 0.10$ are pruned.

3.3.2 PC Algorithm Integration

Constraint-based discovery uses `causal-learn` [18]’s `pc` function with the Fisher-z test at $\alpha = 0.05$. Undirected edges in the resulting CPDAG are resolved by orienting them from the lower to the higher feature index.

Cyclic edges introduced by this step are detected and removed in two phases: first, Kahn’s algorithm checks whether a cycle exists by iteratively stripping zero-in-degree

nodes and comparing the visited count against n ; if a cycle is confirmed, a depth-first search removes the minimal set of back edges needed to restore acyclicity, where a back edge is any edge $i \rightarrow j$ encountered while j is still on the active DFS stack.

3.3.3 $X \rightarrow Y$ Edge Augmentation Rule

Both PC and LiNGAM operate on the X -only feature matrix and therefore produce no edges to Y . Because Shapley Flow requires at least one direct $X_i \rightarrow Y$ edge per feature to accumulate edge credit, an explicit augmentation step appends $X_i \rightarrow Y$ edges for every feature in the prediction model. In practice this connects all features to Y , since feature selection is disabled. The same rule is applied uniformly across all methods and to the True DAG reference graph.

3.3.4 Discovery Performance

Table 2: Causal discovery F1 scores ($X \rightarrow X$ edges only). The discovery-quality ordering of the two algorithms reverses between datasets.

Dataset	Algorithm	F1
Synthetic	PC	0.567
Synthetic	LiNGAM	0.250
Sachs	PC	0.167
Sachs	LiNGAM	0.326

On the synthetic track, PC outperforms LiNGAM (F1: 0.567 vs. 0.250): conditional independence tests partially block confounder-induced associations, while LiNGAM’s functional causal model conflates them with direct paths, flooding the graph with 81 false positives. On Sachs the ordering flips (F1: 0.326 vs. 0.167), log-transformed protein concentrations retain non-Gaussian residuals that LiNGAM can exploit, a property that PC’s Fisher-z test cannot leverage. Notably, both algorithms perform substantially worse on the real data despite the larger sample size, and PC’s output required removing two cycles before it was a valid DAG, a sign that real biological signal is harder to recover than synthetic confounded structure.

3.4 Shapley Computation Settings

Computing exact Shapley values requires evaluating every possible coalition of features, and for a 50-feature system this amounts to 2^{50} coalitions. The cost is severe even for the graph-free baseline, but it is compounded for the structure-aware methods: because they follow the DAG during the value computation, sampling interventional values

along a topological order or activating edges in sequence, each coalition evaluation is itself more expensive than a single model call, so an exhaustive enumeration becomes outright unfeasible in practice. For this reason every Shapley method in this pipeline, the graph-free baseline included, is computed with the same Monte Carlo permutation approximation rather than by exact enumeration, using $T = 100$ sampled permutations. For each dataset, 30% of training rows serve as the background reference distribution (240 rows for synthetic; 1,791 for Sachs), and up to 100 test instances are explained. All random operations use seed 42.

Ten combinations of structure-aware Shapley method and causal graph are computed for the synthetic dataset: Traditional Shapley (baseline) plus the three structure-aware methods applied to three graph sources (PC, LiNGAM, True DAG). The Sachs dataset mirrors this design: Traditional Shapley plus the three methods applied to PC, LiNGAM, and the consensus reference DAG, which here serves the same oracle role that the True DAG plays on the synthetic track. The consensus Sachs graph is thus used both for evaluating discovery quality (Table 2) and as the oracle Shapley input against which the discovered-graph attributions are compared. The same $T = 100$ permutation budget is applied to every combination on both tracks.

Table 3 summarizes what each method changes relative to Traditional Shapley. It is designed to be read one row at a time: each structure-aware method modifies exactly one ingredient of the Traditional recipe. Asymmetric Shapley changes only the permutation space; Causal Shapley changes only the value function, with the ordering following from the components; and Shapley Flow changes who the players are.

Table 3: What Each Method Changes Relative to Traditional Shapley.

Property	Traditional	Asymmetric	Causal	Flow
Players	Features	Features	Features	Edges
Permutation space	All $N!$ orderings	Topological orderings of feature DAG	Topological orderings of component DAG	All orderings of edge set
Value function	Obs. $\mathbb{E}[f_{X_S=x_S}]$	Observational (same)	Interventional $\mathbb{E}[f_{\text{do}(X_S=x_S)}]$	Foreground/background by edge activation
Graph’s role	None	Constrains ordering	Defines components, ordering, and intervention	Supplies the players (edges)

3.4.1 Traditional Shapley Values (Graph-Free Baseline)

The baseline discards all causal information. All features are treated symmetrically: $T = 100$ uniformly random permutations of all features are sampled and, building each coalition incrementally, the marginal contribution of each feature is computed by replacing the absent features with values drawn from the background data. This is the standard SHAP Monte Carlo estimator [7] and is the graph-free reference against which all structure-aware methods are benchmarked.

The coalition value $v(S) = \mathbb{E}[f(X) \mid X_S = x_S]$ is estimated by the `COALITION_VALUE` subroutine (Appendix 5.3), which overwrites the coalition columns of the background matrix with the instance’s real values and averages the model output, and which is reused unchanged by Asymmetric Shapley.

Algorithm 1 Traditional Shapley (Monte Carlo)

INPUT: instance x , model f , background data D , number of permutations T

OUTPUT: attribution vector $\phi \in \mathbb{R}^n$

```

1:  $\phi \leftarrow \mathbf{0}_n$ 
2: baseline  $\leftarrow \text{mean}(f(D))$  ▷  $v(\emptyset)$ : all features marginalised
3: for  $t = 1 \dots T$  do
4:    $\pi \leftarrow \text{RANDOM\_PERMUTATION}(1 \dots n)$  ▷ unconstrained uniform ordering
5:    $S \leftarrow \emptyset$ ,  $v_{\text{prev}} \leftarrow \text{baseline}$ 
6:   for  $i$  in  $\pi$  do ▷ add features one at a time
7:      $S \leftarrow S \cup \{i\}$ 
8:      $v_{\text{curr}} \leftarrow \text{COALITION\_VALUE}(x, S, f, D)$ 
9:      $\phi[i] \leftarrow \phi[i] + (v_{\text{curr}} - v_{\text{prev}})$ 
10:     $v_{\text{prev}} \leftarrow v_{\text{curr}}$ 
11:   end for
12: end for return  $\phi / T$ 

```

The graph is never consulted, which is the defining property of the method. The cost is $O(T \cdot n \cdot |D|)$ model rows evaluated per instance.

3.4.2 Asymmetric Shapley Values

The Asymmetric Shapley implementation retains the same observational coalition value as Traditional Shapley but replaces the uniform random permutation with a randomized Kahn topological sort that draws uniformly from the linear extensions of the feature DAG (the `SAMPLE_TOPOLOGICAL_ORDERING` subroutine, Appendix 5.3). The modification is exclusively in the permutation space; the coalition values are computed identically through `COALITION_VALUE`. The children lists and initial in-degrees are precomputed

once over the X -only subgraph, since Y is never a player; if the feature DAG contains a cycle, the constraints are disabled and the method reduces to Traditional Shapley.

Algorithm 2 Asymmetric Shapley

INPUT: instance x , model f , background data D , feature DAG G , permutations T

OUTPUT: attribution vector $\phi \in \mathbb{R}^n$

PRECOMPUTE: `children[]`, `in_degree[]` from the X -only edges of G (Subroutine 6)

```

1:  $\phi \leftarrow \mathbf{0}_n$ , baseline  $\leftarrow \text{mean}(f(D))$ 
2: for  $t = 1 \dots T$  do
3:    $\pi \leftarrow \text{SAMPLE\_TOPOLOGICAL\_ORDERING}()$   $\triangleright$  ancestors precede descendants
4:    $S \leftarrow \emptyset$ ,  $v_{\text{prev}} \leftarrow \text{baseline}$ 
5:   for  $i$  in  $\pi$  do
6:      $S \leftarrow S \cup \{i\}$ 
7:      $v_{\text{curr}} \leftarrow \text{COALITION\_VALUE}(x, S, f, D)$   $\triangleright$  same value fn as Traditional
8:      $\phi[i] \leftarrow \phi[i] + (v_{\text{curr}} - v_{\text{prev}})$ 
9:      $v_{\text{prev}} \leftarrow v_{\text{curr}}$ 
10:  end for
11: end for return  $\phi / T$ 

```

The cost is the same order as Traditional Shapley; the topological sampler adds only $O(n)$ per permutation. The graph is consulted solely to build the ordering structures at initialization, and the value function never sees it.

3.4.3 Causal Shapley Values

Causal Shapley replaces the observational coalition value with post-interventional sampling that approximates Pearl’s do-operator, $v_{\text{do}}(S) = \mathbb{E}[f(X) \mid \text{do}(X_S = x_S)]$. The method has two nested levels of stochasticity that must be kept distinct: an outer loop that draws a fresh uniform linear extension of the component DAG for each Shapley trial, and an inner loop that, for each coalition, estimates the interventional value from M draws of the post-interventional distribution while traversing components in a fixed deterministic topological order so that each node’s parents are resolved before the node itself. Each draw fixes the intervened features and fills in the remaining features from their parents using the closed-form conditional Gaussian of the background data; inside a confounded component the missing features are drawn independently given their parents (which destroys the spurious within-component correlation), while in an ordinary component they are drawn jointly given the parents and any fixed siblings. In this pipeline the confounder list is empty, so every feature is its own component, the outer ordering reduces to a uniform linear extension of the full feature DAG, and

the inner sampler always takes the univariate-Gaussian branch. The experiments use $T = 100$ outer permutations and $M = 10$ inner samples.

Algorithm 3 Causal Shapley (post-interventional)

INPUT: instance x , model f , background data D , feature adjacency A (X only),
confounder list, outer permutations T , inner samples M

OUTPUT: attribution vector $\phi \in \mathbb{R}^n$

PRECOMPUTE: `components`, `confounded[]`, `parents[]` $\leftarrow$
`BUILD_COMPONENTS( $A$ , confounder list)`

- 1: $\mu \leftarrow \text{mean}(D)$; $\Sigma \leftarrow \text{cov}(D) + \varepsilon I$ $\triangleright \varepsilon$ for numerical stability
- 2: $\phi \leftarrow \mathbf{0}_n$, `baseline` $\leftarrow \text{mean}(f(D))$
- 3: **for** $t = 1 \dots T$ **do** $\triangleright$ OUTER: re-randomised
- 4: $\pi \leftarrow \text{SAMPLE_COMPONENT_TOPO_ORDERING}()$, then expand to features
- 5: $S \leftarrow \emptyset$, $v_{\text{prev}} \leftarrow \text{baseline}$
- 6: **for** i **in** π **do**
- 7: $S \leftarrow S \cup \{i\}$
- 8: $v_{\text{curr}} \leftarrow 0$ $\triangleright$ INNER: estimate v_{do}
- 9: **for** $m = 1 \dots M$ **do**
- 10: $v_{\text{curr}} \leftarrow v_{\text{curr}} + f(\text{POST_INTERVENTIONAL_SAMPLE}(x, S, \dots))$
- 11: **end for**
- 12: $v_{\text{curr}} \leftarrow v_{\text{curr}} / M$
- 13: $\phi[i] \leftarrow \phi[i] + (v_{\text{curr}} - v_{\text{prev}})$
- 14: $v_{\text{prev}} \leftarrow v_{\text{curr}}$
- 15: **end for**
- 16: **end for** **return** ϕ / T

The `POST_INTERVENTIONAL_SAMPLE` subroutine that draws a single sample from $P(X \mid \text{do}(X_S = x_S))$, together with the closed-form conditional-Gaussian draw it relies on, is given in Appendix 5.3.

The cost is $O(T \cdot n \cdot M)$ model evaluations, markedly heavier than Traditional or Asymmetric Shapley because of the inner sampling loop. The graph is used in three places: the directed edges define the parent sets for the interventional draws, the edges plus the confounder list define the component partition and the component DAG that constrains the outer ordering, and the same DAG fixes the deterministic inner traversal order.

3.4.4 Shapley Flow

Shapley Flow treats directed edges as the players of the cooperative game, permuting the full edge set rather than the feature set. In each of $T = 100$ trials all edges ($X \rightarrow X$ and $X \rightarrow Y$) are randomly permuted and activated sequentially, and the system value is evaluated after each activation. The node-value assignment is binary: a node takes its foreground value if it is a source with an active outgoing edge, or if it has an active incoming edge, or if its direct edge to Y is active; otherwise it takes its background value, and Y is always dropped before the model is evaluated. No intermediate model calls are made between X features; the model is evaluated only on the complete node-value vector at each edge addition. Node-level attributions are recovered by summing each node's outgoing edge credits, and by construction the credits satisfy efficiency, summing to $f(x_{\text{fg}}) - f(x_{\text{bg}})$. The $X \rightarrow Y$ augmentation of Section 3 is essential here, as it guarantees every feature has a direct route to accumulate edge credit.

Algorithm 4 Shapley Flow (uniform edge-permutation)

INPUT: foreground instance x_{fg} , background row x_{bg} ,
graph adjacency ($n + 1$ nodes incl. Y), target index Y , trials T

OUTPUT: node attribution vector $\phi \in \mathbb{R}^n$ (Y excluded)

- 1: $E \leftarrow$ list of all directed edges ($u \rightarrow v$) in the graph
- 2: $\text{edge_credit} \leftarrow \{e : 0.0 \mid e \in E\}$
- 3: **for** $t = 1 \dots T$ **do**
- 4: $\text{perm} \leftarrow \text{RANDOM_PERMUTATION}(E)$ $\triangleright$ unconstrained over ALL edges
- 5: $v_{\text{prev}} \leftarrow \text{SYSTEM_VALUE}(\emptyset, x_{\text{fg}}, x_{\text{bg}})$ $\triangleright = f(x_{\text{bg}})$
- 6: $\text{active} \leftarrow []$
- 7: **for** e **in** perm **do**
- 8: $\text{active.append}(e)$
- 9: $v_{\text{curr}} \leftarrow \text{SYSTEM_VALUE}(\text{active}, x_{\text{fg}}, x_{\text{bg}})$
- 10: $\text{edge_credit}[e] \leftarrow \text{edge_credit}[e] + (v_{\text{curr}} - v_{\text{prev}})$
- 11: $v_{\text{prev}} \leftarrow v_{\text{curr}}$
- 12: **end for**
- 13: **end for**
- 14: **for** $e \in E$ **do** $\text{edge_credit}[e] \leftarrow \text{edge_credit}[e] / T$
- 15: **end for**
- 16: $\phi \leftarrow \mathbf{0}_n$ $\triangleright$ aggregate edges $\rightarrow$ features
- 17: **for each** edge ($u \rightarrow v$) $\in E$ **do**
- 18: **if** $u \neq Y$ **then** $\phi[u] \leftarrow \phi[u] + \text{edge_credit}[(u \rightarrow v)]$
- 19: **end if**
- 20: **end for** **return** ϕ

The `SYSTEM_VALUE` subroutine that maps a set of active edges to a model prediction, by assigning each node its foreground or background value under the activation rule, is given in Appendix 5.3.

The cost is $T \cdot (|E| + 1)$ model evaluations per instance (for the synthetic system, roughly $50 \times 276 \approx 1.4 \times 10^4$). This is the most structure-dependent of the four methods: the graph supplies the players themselves, so with no graph there is no game to play.

3.5 Evaluation Metrics

3.5.1 Notation

The following notation is used throughout the evaluation framework. Let N be the number of input features, $\mathcal{F} = \{1, \dots, N\}$ the feature index set, and $\mathcal{I}$ the set of test instances. The trained predictive model is $f : \mathbb{R}^N \rightarrow \mathbb{R}$, and $\hat{\sigma} = \text{std}_{i \in \mathcal{I}} f(x_i)$ is the standard deviation of its predictions over the test set, the natural scale of the attribution space, so every magnitude number in this chapter is reported as a fraction of it. Let $m \in \{\text{Traditional, Asymmetric, Causal, Flow}\}$ index the Shapley method and $G \in \{\emptyset, \text{PC, LiNGAM, True DAG}\}$ the causal graph source.

The per-instance per-feature attribution $\phi_{i,f}^{m,G}$ is the Shapley value assigned to feature $f \in \mathcal{F}$ for instance $i \in \mathcal{I}$ by method m using graph G . Every comparison fixes a subject (m, G) with $G \in \{\text{PC, LiNGAM}\}$ and contrasts it against one of three reference configurations: the **baseline** ref (Traditional Shapley, $G = \emptyset$), the **oracle** ref (True DAG on synthetic, consensus DAG on Sachs), and the **cross-discovery** ref (same method on the opposite discovered graph, $\text{PC} \leftrightarrow \text{LiNGAM}$).

3.5.2 Two Dimensions, Two Metrics, Three Comparisons

Every comparison in this chapter asks the same two questions of a (method, graph) result against a reference: did the attribution change in **magnitude**, and did it change in **direction**? These are the two evaluation dimensions, and each is measured by a single metric used identically in all three comparisons:

- **Magnitude Divergence** (ΔM): how much the absolute attribution diverges from the reference, expressed as a fraction of $\hat{\sigma}$. $\Delta M \geq 0$; $\Delta M = 0.10$ means the typical attribution diverges by 10% of $\hat{\sigma}$.
- **Sign Disagreement** (D): the fraction of (instance, feature) attributions that point in the opposite direction to the reference. $D \in [0, 1]$; $D = 0$ means perfect directional agreement.

The only thing that changes between comparisons is the reference, which is carried as a tag so the same two names cover all six cells of Table 4:

- Tag **base** — reference is Traditional Shapley (how far a graph moves attributions away from the graph-free baseline); Section 4.2.
- Tag **disc** — reference is the opposite discovered graph (how much the choice of discovery algorithm alone perturbs attributions); Section 4.3.
- Tag **oracle** — reference is the True / consensus DAG (how faithfully a discovered graph recovers the oracle attributions); Section 4.4.

Table 4: Evaluation Metric Framework. Two metrics, two dimensions, three reference comparisons.

Dimension	vs. Traditional (base)	vs. Oracle (oracle)	DAG PC vs. LiNGAM (disc)
Magnitude	ΔM_{base}	ΔM_{oracle}	ΔM_{disc}
Sign / Dir	D_{base}	D_{oracle}	D_{disc}

3.5.3 Three Measurement Levels

Both metrics are defined once at the atomic instance level and then lifted to the feature and global levels by fixed aggregation operators, so a single definition serves all three reporting granularities. The level is stated wherever a value is reported; as a convention, tables give the global level, heatmaps the feature level, and scatter plots the instance level.

Instance level (signed). The atomic magnitude quantity is the signed gap between absolute attributions for one (instance, feature):

$$\delta_{i,f}^{\text{ref}} = \left| \phi_{i,f}^{m,G} \right| - \left| \phi_{i,f}^{\text{ref}} \right|. \quad (6)$$

This keeps its sign on purpose: a positive value means the discovered graph inflates the feature’s local importance relative to the reference, a negative value means it suppresses it. The atomic sign quantity is the disagreement indicator $\mathbf{1}[\text{sign}(\phi_{i,f}^{m,G}) \neq \text{sign}(\phi_{i,f}^{\text{ref}})]$, defined only on instances where the reference attribution is non-zero.

Feature level (non-negative). Per feature, the instance gaps are collapsed by a root-mean-square over instances and normalised by $\hat{\sigma}$:

$$\Delta M_f^{\text{ref}} = \frac{1}{\hat{\sigma}} \sqrt{\frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} (\delta_{i,f}^{\text{ref}})^2}. \quad (7)$$

The RMS (rather than a plain mean) measures the typical size of the per-instance change without letting positive and negative gaps cancel, and dividing by $\hat{\sigma}$ puts the

result on a single interpretable scale (percentage of model-output standard deviation) that is comparable across features and across datasets of different raw units.

The choice of $\hat{\sigma}$ as that scale is not arbitrary; it is the natural unit of the attribution space itself. By the efficiency axiom, every Shapley method here satisfies $\sum_f \phi_{i,f}^{m,G} = f(x_i) - \mathbb{E}[f(x)]$, so the attributions live on the same scale as the model output and $\hat{\sigma} = \text{std}_i f(x_i)$ is the total dispersion the whole SHAP space has to distribute. Normalising by $\hat{\sigma}$ expresses a magnitude change as a fraction of the attribution budget that actually exists, a value of 0.10 means “a tenth of the model’s explainable variation” regardless of whether the target is a unitless synthetic variable or a raw protein concentration.

The feature-level sign metric is the disagreement rate D_f^{ref} , the fraction of valid instances whose sign disagrees with the reference.

Global level. Both feature-level metrics are averaged over features:

$$\Delta M^{\text{ref}} = \frac{1}{N} \sum_{f \in \mathcal{F}} \Delta M_f^{\text{ref}}, \quad D^{\text{ref}} = \frac{1}{N} \sum_{f \in \mathcal{F}} D_f^{\text{ref}}. \quad (8)$$

The cross-discovery metrics ΔM_{disc} and D_{disc} are the same constructions with the subject fixed to PC and the reference to LiNGAM.

4 Results and Empirical Analysis

4.1 Causal Discovery Baseline Performance

4.1.1 Performance on the Confounded Linear System

The five hidden confounders introduce spurious correlations that directly violate the causal sufficiency assumption of both estimators. PC exhibits high structural resilience, recovering 40 of 87 true $X \rightarrow X$ edges with precision 0.741 and F1 of 0.567. Thanks to conditional independence testing evaluates localized relationships, it can partially block extended confounder-induced association paths through careful conditioning on intermediate variables [15], allowing PC to maintain a structurally conservative graph relatively close to the true sparse structure.

DirectLiNGAM experiences severe structural degradation, reporting 105 edges with only 24 correct and 81 false positives with precision 0.23 and F1 of 0.25. Standard linear non-Gaussian functional models are highly sensitive to unmeasured variables; because latent confounders violate the foundational assumption of mutually independent exogenous noise, the algorithm frequently misinterprets confounder-induced correlations as direct causal pathways [6, 14]. This over-discovery creates the central experimental

contrast: the same Shapley methods receive completely different structural priors from the two discovery algorithms.

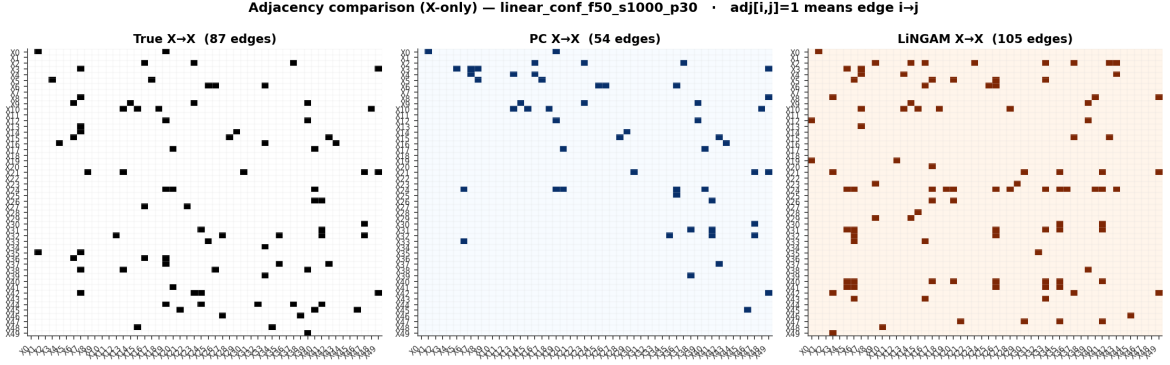


Figure 2: $X \rightarrow X$ adjacency matrices for the synthetic linear-confounded dataset, where $\text{adj}[i, j] = 1$ denotes an edge $i \rightarrow j$. The True DAG (87 edges) is shown alongside the PC (54 edges) and LiNGAM (105 edges) recoveries. PC stays close to the sparse true structure, whereas LiNGAM scatters spurious edges across the matrix, the visual signature of the over-connection that drives its low precision.

4.1.2 Performance Reversal on the Sachs Cell Signaling Dataset

On the real-world Sachs dataset, the performance hierarchy reverses: LiNGAM outperforms PC ($F1 = 0.326$ vs. 0.167 ; precision = 0.269 vs. 0.158). Real intracellular signaling pathways generate joint distributions with strong non-Gaussian marginals even after $\log_{1p}$ transformation, a property that LiNGAM’s functional causal model is explicitly designed to exploit. Conversely, PC’s Fisher-z test assumes linear, multivariate Gaussian residuals. This introduces a known sample-size paradox: while larger sample sizes generally improve causal discovery, providing PC with a massive biological dataset (5,972 training rows compared to 800 in the synthetic side) gives the statistical tests so much power that they become hypersensitive to minor non-Gaussian distributional violations [4, 10]. Consequently, PC loses its discriminative thresholding power under this physical complexity, leading to severe under-recovery.

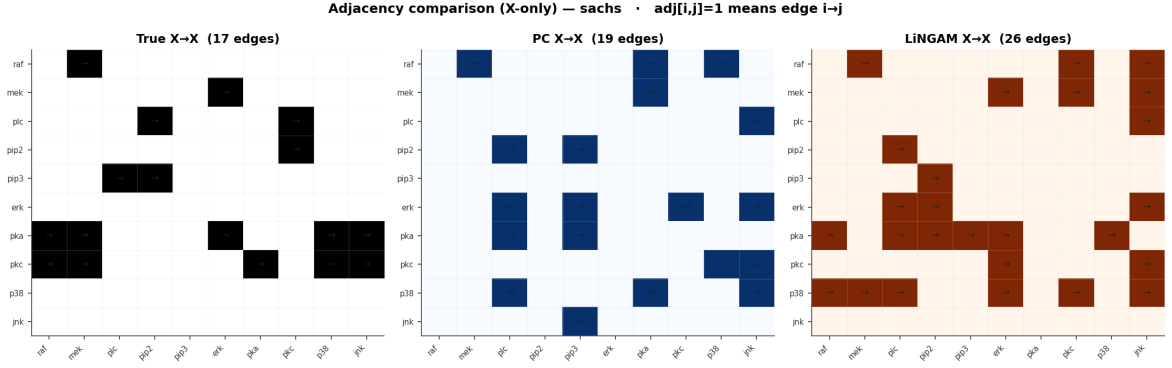


Figure 3: $X \rightarrow X$ adjacency matrices for the Sachs dataset, where $\text{adj}[i, j] = 1$ denotes an edge $i \rightarrow j$. The consensus reference DAG (17 edges) is compared against the PC (19 edges) and LiNGAM (26 edges) recoveries over the ten signaling proteins. Both algorithms recover only a fraction of the true links and introduce edges absent from the consensus network, with LiNGAM the denser of the two.

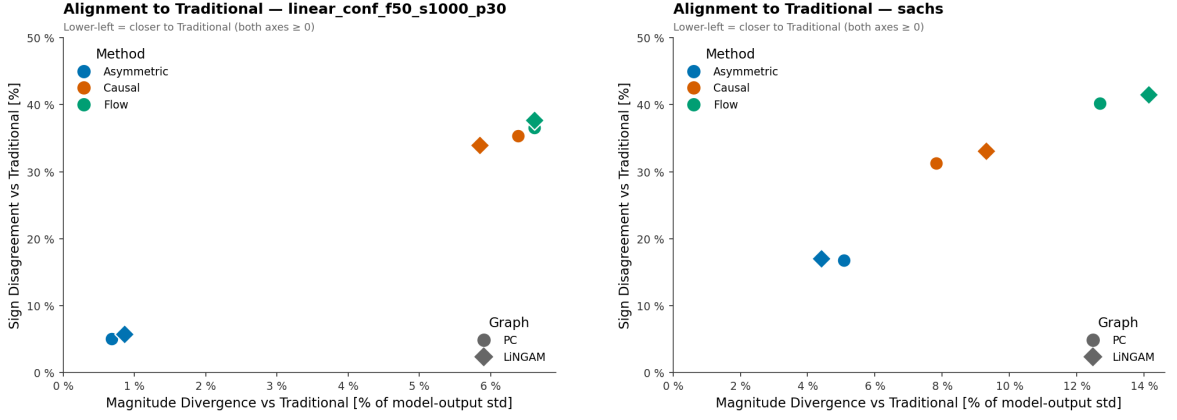
4.2 Graph-Free Baseline Deviation Analysis

Following the metric definitions of Section 3.5, this evaluation block compares each structure-aware method and discovered-graph combination against the Traditional Shapley baseline ($\Delta M_{\text{base}}, D_{\text{base}}$) to verify the expected theoretical behaviour of each method and validate that this behaviour holds on real data.

The theoretical ordering of sensitivity to causal graph injection is given directly by Table 3. Asymmetric Shapley is the least conditioned by the graph: it uses the discovered edges only to restrict the permutation space while keeping the observational value function unchanged, so its deviation from the graph-free baseline depends entirely on how many orderings the graph forbids. Causal Shapley is more sensitive: the graph enters the value function itself, driving post-interventional distributions for features outside the coalition via their discovered parent sets, so every false or missing edge corrupts an attribution directly. Shapley Flow is the most sensitive: the discovered edges are the players of the game, and the entire credit-routing mechanism collapses without them, so any graph error propagates across all downstream paths simultaneously.

Two cross-cutting patterns emerge from Tables 5–6 and are visible immediately in the scatter plots. First, Sign Disagreement D_{base} is a more severe perturbation than Magnitude Divergence ΔM_{base} : introducing a causal graph flips the direction of a larger share of feature attributions than it inflates their absolute size. This is because sign inversions arise whenever a moderate per-feature change crosses zero, a threshold many near-zero attributions sit close to, while large magnitude changes are concentrated in a handful of dominant features. Second, the choice of discovered graph, PC or LiNGAM, matters little for the graph-versus-no-graph contrast: both discovered graphs perturb

ΔM_{base} and D_{base} by nearly identical amounts for a given method, so the dominant driver of deviation is the method’s structural coupling to any causal graph, not the specific graph supplied.



(a) Synthetic dataset. Asymmetric Shapley clusters tightly in the lower-left corner under both graphs, while Causal and Flow sit far to the upper-right, deviating strongly on both axes regardless of which graph supplies the structure.

(b) Sachs dataset. The same lower-left clustering of Asymmetric Shapley holds, but both axes spread wider than on synthetic and the PC/LiNGAM markers separate more visibly for Causal and Flow, but still less than 2% apart, the early signal of the discovery-algorithm sensitivity examined in Section 4.3.

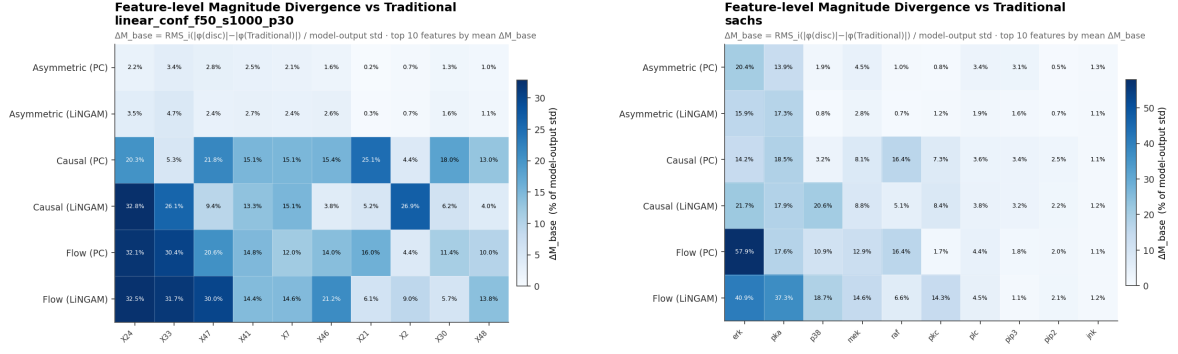
Figure 4: Magnitude versus sign deviation from Traditional Shapley.

Table 5: Baseline Deviation – Linear-Conf Synthetic Dataset.

Method	Graph	D_{base}	ΔM_{base}
Asymmetric	PC	5.08%	0.68%
Asymmetric	LiNGAM	5.76%	0.86%
Causal	PC	35.30%	6.39%
Causal	LiNGAM	33.92%	5.84%
Flow	PC	36.58%	6.62%
Flow	LiNGAM	37.68%	6.61%

Table 6: Baseline Deviation – Sachs Cell Signaling Dataset.

Method	Graph	D_{base}	ΔM_{base}
Asymmetric	PC	16.80%	5.08%
Asymmetric	LiNGAM	17.00%	4.41%
Causal	PC	31.30%	7.83%
Causal	LiNGAM	33.10%	9.31%
Flow	PC	40.20%	12.68%
Flow	LiNGAM	41.50%	14.17%

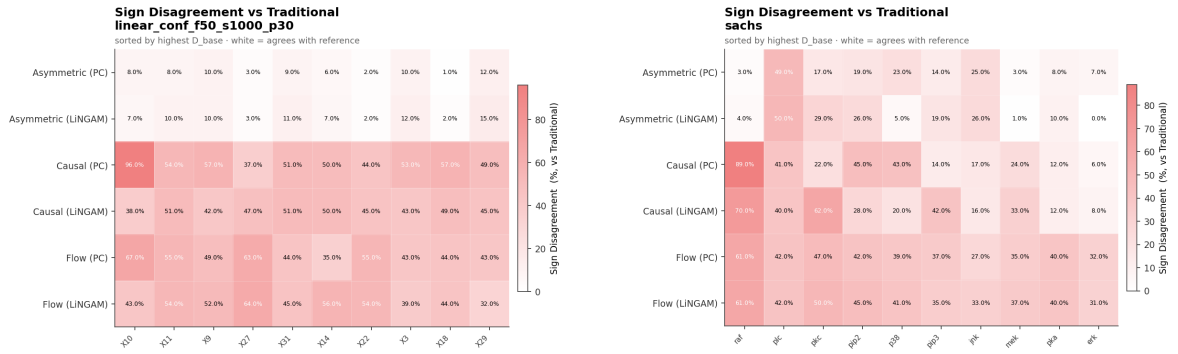


(a) Synthetic dataset (top 10 features by mean ΔM_{base}). Rows are method-graph combinations; columns are features. The Asymmetric rows are near-zero throughout, while Causal and Flow concentrate their largest deviations on X24, X33 and X47.

(b) Sachs dataset (top 10 features by mean ΔM_{base}). The deviation concentrates on erk and pka, with Flow under PC producing the single largest feature-level divergence against Traditional Shapley (erk, 57.9% of $\hat{\sigma}$).

Figure 5: Feature-level Magnitude Divergence ΔM_{base} vs. Traditional Shapley.

The feature-level magnitude deviations summarized by the global ΔM_{base} are shown in full in Figures 5, which lay out the per-feature ΔM_{base} against Traditional Shapley for the top perturbed features of each dataset across all six method-graph combinations. The heatmaps make the method hierarchy visually immediate: the two Asymmetric rows are almost uniformly pale, while the Causal and Flow rows darken sharply on the highest-importance features (X24, X33, X47 on synthetic; erk, pka on Sachs), confirming that the magnitude divergence concentrates on a small set of dominant features and is driven by the interventional and edge-routing methods rather than by Asymmetric.



(a) Synthetic dataset (top 10 features by mean D_{base}). Rows are method-graph combinations; columns are features. The Asymmetric rows are near-zero throughout; Causal and Flow show elevated sign-flip rates concentrated on the same dominant features as the magnitude heatmap.

(b) Sachs dataset (top 10 features by mean D_{base}). Sign disagreement concentrates on raf, plc, pkc, with Causal under PC producing the highest per-feature sign-flip rate against the graph-free baseline.

Figure 6: Feature-level Sign Disagreement D_{base} vs. Traditional Shapley.

The Shapley method hierarchy by sign disagreement is similar: Asymmetric rows remain

near-zero throughout, while Causal and Flow show elevated sign-flip rates, but the nodes flagged by the sign heatmap are *not* the same nodes flagged by the magnitude heatmap. On synthetic, the highest sign disagreement concentrates on X10 and X11; on Sachs it concentrates on raf, plc, and pkc. These are different features from the magnitude leaders (X24, X33, X47; erk, pka). The reason is that sign inversions arise when a graph-induced attribution shift is large enough to cross zero, and the features most vulnerable to this are those whose Traditional Shapley values already sit near zero: a modest structural change in the discovered DAG is sufficient to flip their sign without producing a large RMS magnitude divergence. The dominant features, by contrast, carry large attributions that absorb the same redistribution without changing direction. The practical implication is that the sign and magnitude heatmaps are complementary diagnostics, not redundant ones. Elevated D_{base} on a feature is a direct alert to inspect whether the discovered graph’s structural changes for that feature, a new parent, a reversed edge, or a spurious child, provide a causally coherent explanation for the sign flip; if they do not, the graph in that neighbourhood is likely unreliable and should be validated before using any structure-aware method for that node.

4.2.1 Asymmetric Shapley, High Robustness to Graph Injection

Asymmetric Shapley achieves the closest agreement with the Traditional Shapley baseline across both datasets. On the synthetic track, $D_{\text{base}} = 5.08\%$ (PC) and 5.76% (LiNGAM), with $\Delta M_{\text{base}} = 0.68\%$ and 0.86% respectively: fewer than 6% of attribution signs change and the magnitude divergence is under 1% of $\hat{\sigma}$. On Sachs the deviation D_{base} triples (16.80–17.00%), still well below the deviation seen under Causal Shapley and Shapley Flow on the synthetic dataset, and ΔM_{base} reaches 4.41–5.08% of $\hat{\sigma}$, the larger numbers reflecting the compact 10-node network where each ordering constraint binds a larger share of the graph. In both cases ASV remains the method that perturbs the baseline least.

The reason is structural, ASV changes only the permutation space and leaves the observational value function untouched, so its deviation from Traditional depends solely on how many orderings the graph forbids. The discovered graphs forbid very few, none of the PC or LiNGAM graphs form long directed chains, so most feature pairs stay order-free and the admissible permutations remain close to the full $N!$ set. The averaging therefore runs over almost the same orderings as Traditional and the attributions barely move, least of all on the larger, sparser synthetic graph.

A key reason why this buffer is so effective on the synthetic dataset is the vast scale of $50! \approx 3 \times 10^{64}$: even a graph with hundreds of ordering constraints eliminates only a negligible fraction of the available permutations, so the ASV average is virtually indistinguishable from the unconstrained Traditional average. On Sachs the situation is

fundamentally different: the ordering universe is only $10! = 3,628,800$, and each directed edge in the 10-node discovered graph eliminates a proportionally far larger share of valid orderings. This is the direct combinatorial reason why D_{base} and ΔM_{base} triple on Sachs relative to synthetic: the ordering constraint imposed by each discovered edge is $\sim 10^{57}$ times more influential when $N = 10$ than when $N = 50$, regardless of graph quality.

The higher D_{base} on Sachs ($\sim 17\%$ vs $\sim 5\%$ synthetic) is therefore driven by this combinatorial scaling: the smaller ordering universe means each discovered edge carries far more weight in shaping the final attribution average.

4.2.2 Causal Shapley, Intermediate Deviation with Interventional Redistribution

Causal Shapley diverges from Traditional Shapley far more than Asymmetric with a general sign disagreement D_{base} around 33.92–35.30% and ΔM_{base} around 5.84–6.39% on synthetic, with D_{base} 31.30–33.10% and ΔM_{base} 7.83–9.31% of $\hat{\sigma}$ on Sachs. Replacing observational marginalization with do-distributions redistributes credit based on the discovered graph, changing the sign of roughly a third of all attributions.

A notable pattern across both datasets is that the features with the highest sign-flip rates are not the ones with the largest magnitude changes. Features such as X10 (synthetic) and pkc (Sachs) do not rank among the top features by ΔM_{base} , they fall outside the highest-perturbation columns in the heatmap, yet they show the highest sign disagreement. The reason is that their Traditional Shapley values sit close to zero; a small interventional redistribution is sufficient to push the attribution across zero and invert its sign. Larger-magnitude features absorb the same redistribution without crossing zero. The specific features that best exemplify the source-versus-intermediate credit shift are examined in Section 4.5.

4.2.3 Shapley Flow, High Deviation with Sign Instability

Shapley Flow shows the highest Sign Disagreement on the synthetic dataset ($D_{\text{base}} = 36.58\text{--}37.68\%$) and the largest magnitude deviation overall, reaching $\Delta M_{\text{base}} = 14.17\%$ of $\hat{\sigma}$ on Sachs under LiNGAM. The two move together: re-routing credit along edges produces large magnitude divergences, and the larger the magnitude divergence the more often it is large enough to carry an attribution across zero and invert its sign. Flow’s high magnitude deviation is therefore the direct cause of its high sign instability.

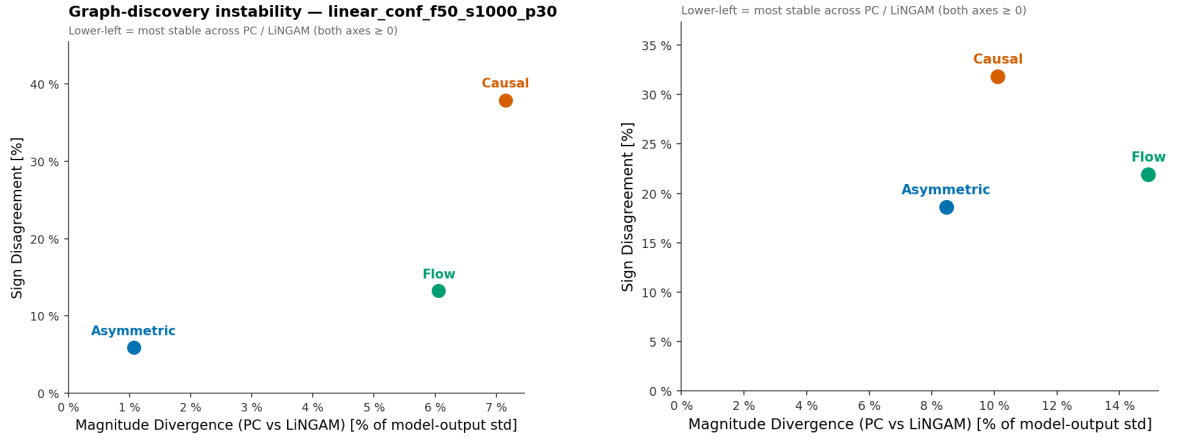
Read per feature, the signed instance-level gap again has a structural meaning, but for Flow the discriminating quantity is the node’s balance of incoming to outgoing edges. A node’s attribution is the sum of its outgoing edge credits, so a feature with many

incoming edges spends its activation propagating its parents’ credit forward and tends to lose magnitude, while a feature with a high outgoing-to-incoming ratio accumulates edge credit and tends to gain it. The features that best illustrate these mechanisms, including cases where a node’s edge balance flips between the two discovered graphs, are deferred to Section 4.5.

4.3 Discovery Algorithm Sensitivity Analysis

Beyond knowing that injecting a causal graph shifts attributions away from the baseline, a practitioner also needs to know how stable those shifts are: if two reasonable discovery algorithms produce different explanations for the same data, the specific graph supplied is itself a source of uncertainty. This section quantifies that instability. If a practitioner runs two discovery algorithms on the same data and feeds each graph into the same Shapley method, how much do the explanations change? The answer is, it depends heavily on the method, and the effect is larger on the sign of attributions than on their magnitude. Asymmetric Shapley consistently shows the lowest sensitivity, it keeps both ΔM_{disc} and D_{disc} well below its counterparts on both datasets. The gap is most visible on the real data, Flow’s magnitude divergence between the two graphs ($\Delta M_{\text{disc}} = 14.92\%$ of $\hat{\sigma}$) is nearly double Asymmetric’s (8.46%), meaning that choosing Shapley Flow instead of Asymmetric makes the explanation twice as sensitive in magnitude to the choice of discovery algorithm. On the sign axis, choosing between Causal Shapley and Asymmetric Shapley for the Sachs dataset determines whether switching from one discovered graph to the other flips 31.80% or 18.60% of attribution signs, Causal disagrees with itself across graphs at almost double the rate of Asymmetric.

Figures 7 plot the two quantities against each other; the lower-left corner marks a method whose attributions are stable regardless of which discovery algorithm was used. Tables 7–8 quantify how much the choice between PC and LiNGAM affects the final attributions for each Shapley method.



(a) Synthetic dataset. Asymmetric sits in the lower-left (stable on both axes), Causal in the upper-right (unstable on both), and Flow in between, magnitude-sensitive but comparatively sign-stable.

(b) Sachs dataset (ΔM_{disc}). The same ordering holds, with both axes wider than on synthetic; Causal remains the most sign-unstable while Flow carries the largest magnitude difference between graphs.

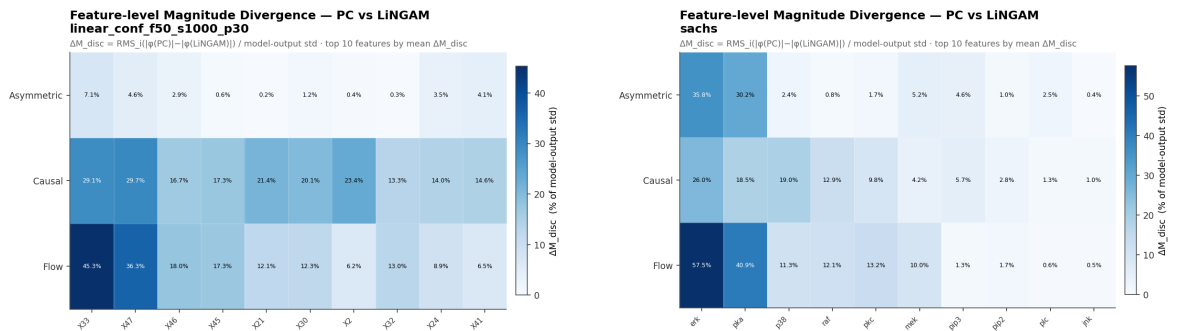
Figure 7: Graph-discovery instability (PC vs. LiNGAM).

Table 7: PC vs. LiNGAM Sensitivity – Synthetic.

Method	ΔM_{disc}	D_{disc}
Asymmetric	1.07%	5.93%
Causal	7.15%	37.94%
Flow	6.06%	13.27%

Table 8: PC vs. LiNGAM Sensitivity – Sachs.

Method	ΔM_{disc}	D_{disc}
Asymmetric	8.46%	18.60%
Causal	10.11%	31.80%
Flow	14.92%	21.88%



(a) Synthetic dataset (top 10 features by mean ΔM_{disc}). The Asymmetric row is near-zero; Causal and Flow concentrate their PC-vs-LiNGAM magnitude differences on X33 and X47, with Flow producing the single darkest cell on X33.

(b) Sachs dataset (top 10 features by mean ΔM_{disc}). The sensitivity concentrates on erk and pka across all three methods, with Flow reaching a feature-level ΔM_{disc} of 57.5% of $\hat{\sigma}$ on erk.

Figure 8: Feature-level cross-discovery Magnitude Divergence ΔM_{disc} (PC vs. LiNGAM).

The feature-level breakdown of ΔM_{disc} reveals which features have contested causal connections, nodes whose discovered neighbourhood differs materially between PC and LiNGAM and whose attributions therefore shift depending on which graph is used. The features flagged by ΔM_{disc} are the same ones already flagged by ΔM_{base} : on synthetic the cross-discovery magnitude instability concentrates on X33 and X47, and on Sachs almost entirely on erk and pka, exactly the dominant nodes that showed the largest deviations from the Traditional Shapley baseline in Section 4.2. This convergence is not coincidental. Both metrics measure how much a feature’s attribution diverges when the causal graph changes: ΔM_{base} captures the divergence when a graph is introduced at all (any discovered graph versus no graph), while ΔM_{disc} captures the divergence when the graph switches between two imperfect alternatives. A feature that ranks high on both lists is one whose Shapley attribution is structurally sensitive in two independent senses, it is affected by graph injection and it is affected by graph choice, pointing to a local causal neighbourhood that is both influential and poorly agreed upon by the discovery algorithms. Used together, ΔM_{base} and ΔM_{disc} at the feature level provide a targeted audit: they narrow the practitioner’s attention from the full feature set to the handful of nodes where structural uncertainty actually drives attribution variance.

On the synthetic dataset, Asymmetric Shapley confirms the sparsity buffer already seen in the oracle comparison: its per-feature ΔM_{disc} stays well within a single percentage point of $\hat{\sigma}$ across all features, meaning that switching between the PC and LiNGAM graphs has essentially no effect on ASV attributions at the individual feature level. The 50-node sparse structure leaves most ordering relationships unconstrained by either discovered graph, so the two variants of ASV average over nearly the same permutation space and produce nearly identical attributions. The picture shifts on Sachs: even Asymmetric shows elevated ΔM_{disc} on erk and pka, reflecting the large structural disagreement between PC and LiNGAM on those proteins in the compact 10-node network. This once again, is not a failure of the method but a consequence of the dataset regime, in low-dimensional graphs, discovery errors and inter-algorithm disagreements bind a large fraction of the available orderings, and ASV cannot average them away. Even so, the ΔM_{disc} feature-level heatmap remains useful in exactly this regime: it clearly identifies erk and pka as the contested nodes, giving the practitioner a precise and actionable target for graph validation rather than a global warning about the entire network.

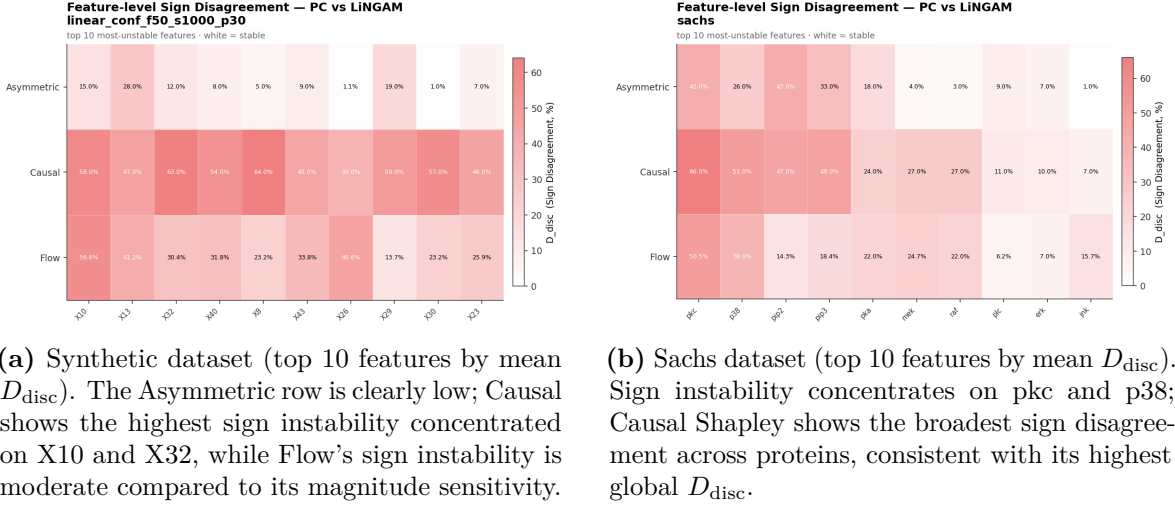


Figure 9: Feature-level cross-discovery Sign Disagreement D_{disc} (PC vs. LiNGAM).

Causal Shapley carries the largest cross-discovery sign instability on both tracks, reaching up to 64% disagreement on individual features, meaning that for 64 out of 100 evaluated instances the attribution sign flips depending solely on which discovery algorithm was used, which is a very high instability. This concentrates across methods on X10 for the synthetic dataset and on pkc for Sachs.

The cross-discovery sign heatmaps also confirm the sparsity buffer for ASV already seen in the magnitude analysis. On the synthetic track, ASV’s row in the D_{disc} heatmap is almost uniformly pale: the high-dimensional sparse graph leaves most ordering relationships unconstrained by both PC and LiNGAM, so the two ASV variants sample nearly identical permutation spaces and produce nearly identical attribution directions. On Sachs, the same buffer weakens for the same reason, and ASV’s sign instability on pkc and p38 is no longer well separated from Causal and Flow.

A pattern that runs through the cross-discovery sign heatmap on both datasets is that the features with the highest D_{disc} are not the same features with the highest ΔM_{disc} . On synthetic, X10 and X32 carry the largest cross-discovery sign disagreement while X33 and X47 carry the largest cross-discovery magnitude divergence; on Sachs, sign instability peaks on pkc and p38 while magnitude instability peaks on erk and pka. The logic is the same as in every earlier sign comparison: sign flips occur where a feature’s attribution sits close to zero, so switching from one discovered graph to the other is enough to cross the zero boundary without leaving a large RMS footprint. Features with already small attributions need only a modest structural change to invert their direction, whereas dominant features absorb the same structural change without changing sign. The D_{disc} heatmap is therefore a complementary tool to ΔM_{disc} : where ΔM_{disc} flags features whose importance is contested between the two algorithms, D_{disc}

flags features whose explainability is inherently fragile, sitting so close to zero that the direction of the explanation is determined by which discovery algorithm was run rather than by any stable structural signal.

4.3.1 Asymmetric Shapley, Minimal Sensitivity on Synthetic

On the synthetic dataset, ASV shows the lowest sensitivity to discovery algorithm choice: $\Delta M_{\text{disc}} = 1.07\%$ of $\hat{\sigma}$, $D_{\text{disc}} = 5.93\%$. Only about 6 in 100 attribution signs differs between the PC and LiNGAM variants on average on all features. This could be explained by the high-dimensional sparse settings of the synthetic dataset, both graphs leave most ordering relationships unconstrained and ASV’s permutation sampling produces nearly identical attribution distributions regardless of which graph is supplied.

On the Sachs dataset, sensitivity increases substantially to $\Delta M_{\text{disc}} = 8.46\%$ of $\hat{\sigma}$ and $D_{\text{disc}} = 18.6\%$. Unlike the synthetic dataset, the Sachs dataset is compact (10 nodes): the different edge sets discovered by PC and LiNGAM impose materially different topological constraints, and these differences accumulate into visible attribution changes, specially in nodes with known influence as *erk* and *pka*.

4.3.2 Causal and Flow, High Sensitivity, Especially on Real Data

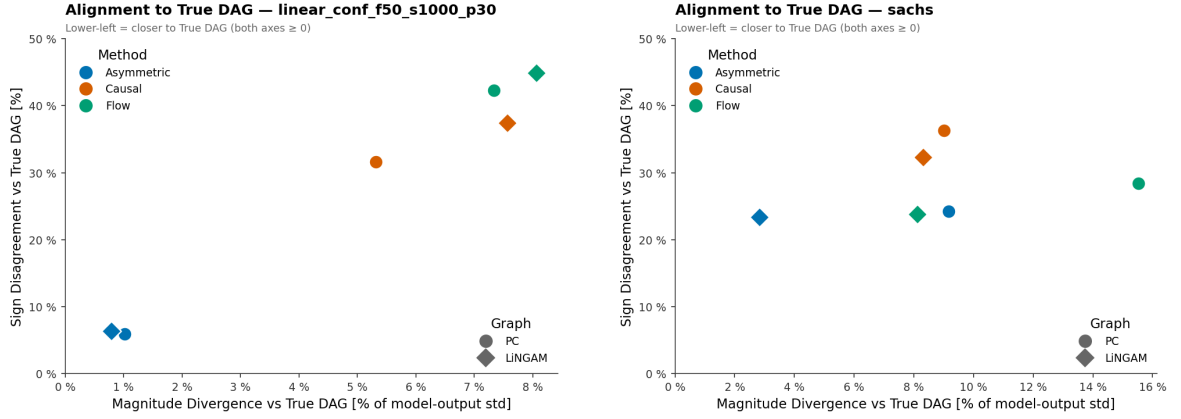
CSV and Shapley Flow are substantially more sensitive to discovery algorithm choice, but in different ways. On the synthetic dataset, CSV’s $\Delta M_{\text{disc}} = 7.15\%$ and Flow’s $\Delta M_{\text{disc}} = 6.06\%$ of $\hat{\sigma}$ are roughly 7 and 6 times larger than ASV’s. The sign axis separates the two methods: CSV’s $D_{\text{disc}} = 37.94\%$ is the highest instability in the synthetic experiment, because interventional conditioning inverts attribution signs whenever the parent sets differ between graphs, which happens frequently given LiNGAM’s 81 spurious edges. Flow, by contrast, is much more sign-stable on synthetic ($D_{\text{disc}} = 13.27\%$) despite its comparable magnitude sensitivity; its cross-discovery sign instability comes largely from both discovered graphs sharing many of the same orientation errors rather than diverging from each other.

On the Sachs dataset the magnitude instability escalates: CSV’s $\Delta M_{\text{disc}} = 10.11\%$ and Flow’s $\Delta M_{\text{disc}} = 14.92\%$ of $\hat{\sigma}$ mean that choosing LiNGAM over PC moves CSV and Flow attributions by more than a tenth of the model-output standard deviation, making the discovery algorithm a dominant source of attribution variance. The sign-disagreement rates converge somewhat (CSV $D_{\text{disc}} = 31.80\%$, Flow 21.88%), with CSV remaining the most sign-unstable method on both tracks. The feature-level heatmaps in Figures 8–9 show that magnitude instability is concentrated on a few nodes: on synthetic, ΔM_{disc} focuses on X33 and X47, and on Sachs almost entirely on *erk* and *pka*, the same dominant nodes that drive every other comparison in this chapter.

Across all three comparison tracks, a consistent pattern holds: magnitude divergence is driven by a handful of focal nodes, which Section 4.5 examines in depth to understand how specific discovery errors interact with each Shapley method. Sign divergence, by contrast, tends to spread more broadly across features, with many near-zero attributions getting pushed across the sign boundary by small magnitude shifts. In both dimensions, Asymmetric Shapley remains the lowest-risk choice when transitioning from a graph-free baseline, whether the discovered graph is noisy, unvalidated, or switches between discovery strategies.

4.4 True-Graph Causal Alignment

The two preceding analyses establish that injecting a causal graph moves attributions away from the graph-free baseline, and that the size of this shift varies systematically depending on which discovery algorithm is used. Neither analysis, however, can say whether the shift moves the explanation closer to or further from the truth. The oracle-alignment analysis now measures how closely each method and discovered-graph combination recovers the attributions that would be obtained with the oracle graph as input. On the synthetic track the oracle is the True DAG; on Sachs it is the consensus reference DAG, which plays the same role. Tables 9–10 report ΔM_{oracle} and D_{oracle} for each track, and Figures 10 place every configuration on the magnitude axis (ΔM_{oracle}) against the sign axis (D_{oracle}), where the lower-left corner marks the closest recovery of the oracle attributions. Better discovery accuracy consistently produces smaller oracle deviation. On the synthetic dataset the margin is modest: Causal Shapley under PC ($F1 = 0.567$) reaches $D_{\text{oracle}} = 31.60\%$ and $\Delta M_{\text{oracle}} = 5.32\%$, versus 37.46% and 7.57% under LiNGAM ($F1 = 0.250$). The effect is more visible on Sachs, where LiNGAM outperforms PC ($F1 = 0.326$ vs. 0.167): Asymmetric Shapley under LiNGAM achieves $\Delta M_{\text{oracle}} = 2.84\%$ versus 9.16% under PC, and Shapley Flow drops from 15.52% to 8.12% when switching from PC to LiNGAM. Among the three methods, Asymmetric Shapley remains the least penalised by graph misspecification, with oracle deviation staying low regardless of which discovered graph is supplied.



(a) Synthetic dataset. Asymmetric Shapley sits in the lower-left corner under both graphs, recovering the oracle attributions almost exactly, while Causal and Flow sit far up and to the right.

(b) Sachs dataset (ΔM_{oracle}). The lower-left ordering is less clean than on the synthetic track: Asymmetric still aligns best on magnitude, but the sign axis compresses the three methods together and the PC/LiNGAM markers separate widely on magnitude.

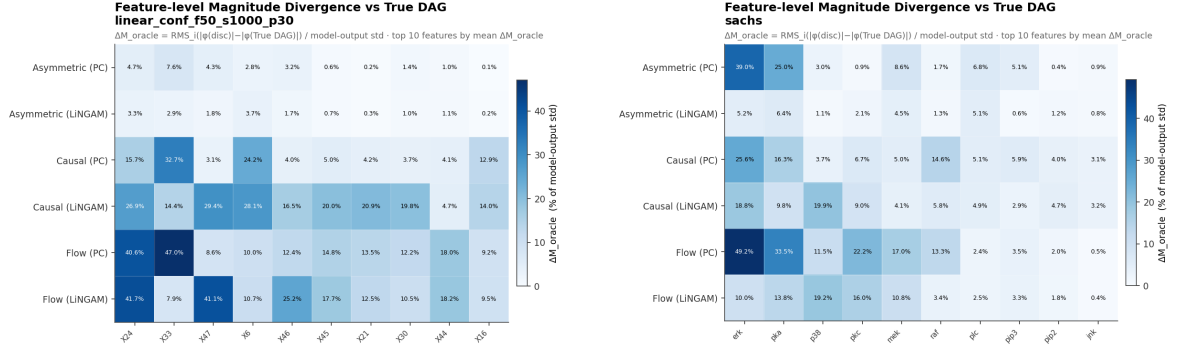
Figure 10: Magnitude versus sign deviation from the oracle DAG.

Table 9: Oracle Alignment – Linear-Conf Synthetic Dataset. D_{oracle} and ΔM_{oracle} in % of $\hat{\sigma}$.

Method	Graph	D_{oracle}	ΔM_{oracle}
Asymmetric	PC	5.88%	1.02%
Asymmetric	LiNGAM	6.30%	0.79%
Causal	PC	31.60%	5.32%
Causal	LiNGAM	37.46%	7.57%
Flow	PC	42.26%	7.34%
Flow	LiNGAM	44.91%	8.06%

Table 10: Oracle Alignment – Sachs Cell Signaling Dataset (oracle = consensus DAG). D_{oracle} and ΔM_{oracle} in % of $\hat{\sigma}$.

Method	Graph	D_{oracle}	ΔM_{oracle}
Asymmetric	PC	24.26%	9.16%
Asymmetric	LiNGAM	23.40%	2.84%
Causal	PC	36.30%	9.01%
Causal	LiNGAM	32.30%	8.31%
Flow	PC	28.36%	15.52%
Flow	LiNGAM	23.81%	8.12%

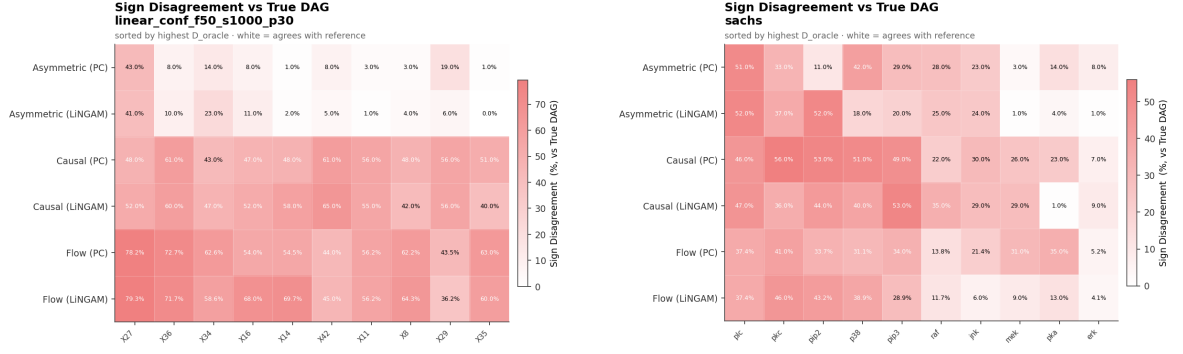


(a) Synthetic dataset (top 10 features by mean ΔM_{oracle}). Asymmetric rows are near-zero; Causal and Flow concentrate their oracle deviations on X24, X33, and X47.

(b) Sachs dataset (top 10 features by mean ΔM_{oracle}). Flow under PC produces the largest single deviation on erk (49.2% of $\hat{\sigma}$), the protein PC isolates by deleting its parent links.

Figure 11: Feature-level Magnitude Divergence ΔM_{oracle} vs. the oracle DAG.

The feature-level breakdown of these oracle deviations reveals a qualitative difference between the two experimental tracks. On the synthetic dataset, the Asymmetric rows stay pale across nearly all features: the 50-node sparse structure buffers ASV against ordering errors because spurious or missing edges affect only a small fraction of the 50! possible orderings, leaving most orderings compatible with both the true and discovered graphs. On Sachs, this protection weakens substantially. With only 10 nodes, each mis-oriented edge eliminates a materially larger fraction of valid topological orderings. The effect concentrates on erk: even Asymmetric under PC produces a feature-level ΔM_{oracle} of roughly 39%, and Flow under PC reaches 49.2% of $\hat{\sigma}$, the worst single deviation in the experiment. The “Asymmetric rows are pale” characterisation therefore holds reliably on large, sparse graphs but breaks down on small, dense networks when discovery errors restructure a node’s local neighbourhood.



(a) Synthetic dataset (top 10 features by mean D_{oracle}). Asymmetric rows are near-zero; Causal and Flow concentrate oracle sign disagreement on X27, X36, and X34, a distinct set from the magnitude leaders (X24, X33, X47), reflecting that near-zero attributions on these features are inverted by relatively small graph-induced shifts.

(b) Sachs dataset (top 10 features by mean D_{oracle}). Sign disagreement concentrates on plc, pkc, and pip2; only pkc overlaps with the magnitude heatmap leaders, confirming that sign-flip-vulnerable features are those whose oracle attributions sit close to zero, not the high-magnitude nodes.

Figure 12: Feature-level Sign Disagreement D_{oracle} vs. the oracle DAG.

Two conclusions emerge from the oracle sign heatmaps. First, ASV’s sparsity buffer holds on the synthetic track, global D_{oracle} stays at roughly 6%, but it fails for X27, which is isolated by both discovery algorithms, pushing ASV’s sign disagreement on that feature to roughly 42%, comparable to Causal and Flow. This finding is only visible with oracle access; in practice the baseline and cross-discovery comparisons serve as the accessible proxies for identifying such structurally fragile features.

Second, on Sachs this sparsity buffering is substantially weaker: the compact 10-node network means that even moderate discovery errors restructure a large fraction of the admissible orderings, and ASV’s D_{oracle} values ($\sim 23\text{--}24\%$) are no longer well separated from those of Causal and Flow ($\sim 23\text{--}36\%$), making it harder to use ASV sign disagreement as a discriminating signal.

Across both tracks, the features that concentrate the highest oracle sign disagreement do not overlap with the magnitude leaders. On synthetic, sign disagreement peaks on X27, X36, and X34 while magnitude deviation peaks on X24, X33, and X47; on Sachs, sign disagreement concentrates on plc, pkc, and pip2 while magnitude deviation concentrates on erk and pka, with only pkc appearing in both lists. This separation is not coincidental: sign flips occur on features whose oracle attributions oscillate near zero, where a graph-induced shift small enough to leave no large RMS footprint is still large enough to cross the sign boundary.

The oracle sign and magnitude heatmaps are complementary diagnostics: ΔM_{oracle} flags features whose importance is structurally over- or under-estimated by the discovered graph, while D_{oracle} flags features sitting close to zero under the oracle and susceptible to directional inversion. This mirrors the complementarity already observed in the

baseline analysis of Section 4.2. Since the oracle comparison is unavailable outside controlled experiments, the baseline $(\Delta M_{\text{base}}, D_{\text{base}})$ and cross-discovery $(\Delta M_{\text{disc}}, D_{\text{disc}})$ comparisons serve as the practical equivalents: together they approximate what the oracle would reveal without requiring a ground-truth graph.

4.4.1 Asymmetric Shapley, Near-Perfect Oracle Fidelity

ASV achieves the highest oracle alignment on both tracks. On synthetic, $D_{\text{oracle}} \approx 6\%$ and $\Delta M_{\text{oracle}} < 1\%$ of $\hat{\sigma}$ regardless of which discovered graph is used. On Sachs, ASV again records the lowest magnitude deviation ($\Delta M_{\text{oracle}} = 2.84\%$ under LiNGAM, the best of any configuration on the real track), though sign disagreement rises to $\sim 24\%$ due to the compact network. Notably, graph quality affects the sign and magnitude axes differently: D_{oracle} is essentially tied between PC and LiNGAM on both datasets, but the more accurate graph does reduce magnitude deviation, most clearly on Sachs, where LiNGAM achieves a threefold improvement over PC.

4.4.2 Causal and Flow, Compounding Error Under Imprecise Graphs

CSV and Flow are substantially more sensitive to graph quality. On synthetic, CSV reaches $D_{\text{oracle}} = 31.60\text{--}37.46\%$ and Flow $D_{\text{oracle}} = 42.26\text{--}44.91\%$, meaning close to half of all Flow attributions point in the wrong direction relative to the True DAG. On Sachs both methods improve with the better LiNGAM graph: CSV drops from 36.30% to 32.30% and Flow’s magnitude deviation falls from 15.52% to 8.12% of $\hat{\sigma}$. Flow’s high synthetic sign instability is partly a property of the dense, heavily mis-oriented graphs on that track rather than an invariant of the method.

4.5 Granular Case Studies

Four archetypal error mechanisms illustrate the micro-level behaviour observed across both experimental datasets. In each case the starting point is the same workflow a practitioner without a ground-truth graph would follow: elevated global divergence $(\Delta M, D)$ flags that a method’s attributions are moving substantially; feature-level heatmaps for the graph-free $(\Delta M_{\text{base}}, D_{\text{base}})$ and cross-discovery $(\Delta M_{\text{disc}}, D_{\text{disc}})$ comparisons narrow attention to the specific nodes driving that movement; and inspecting the discovered neighbourhood of those nodes reveals the structural error responsible. The True DAG (or the Sachs consensus DAG) is then used only to confirm that the observable signals were pointing at the right place, it plays the role of a post-hoc validator, not a diagnostic input.

4.5.1 Correlated and Contested Discovery Failures

Interventional frameworks (CSV) and edge-routing models (Flow) over-inflate the attribution of any node that a discovered graph misrepresents as a high-degree root source. The clearest synthetic case is X24, a true direct parent of Y with 3 incoming and 4 outgoing edges in the True DAG. Both PC and LiNGAM strip its true parents and add spurious children (Figure 13), recasting it as an apparent high out-degree source; this inflates its interventional parent set under CSV and its outgoing-edge credit under Flow, so it absorbs compounded attribution from paths that do not exist in the true graph.

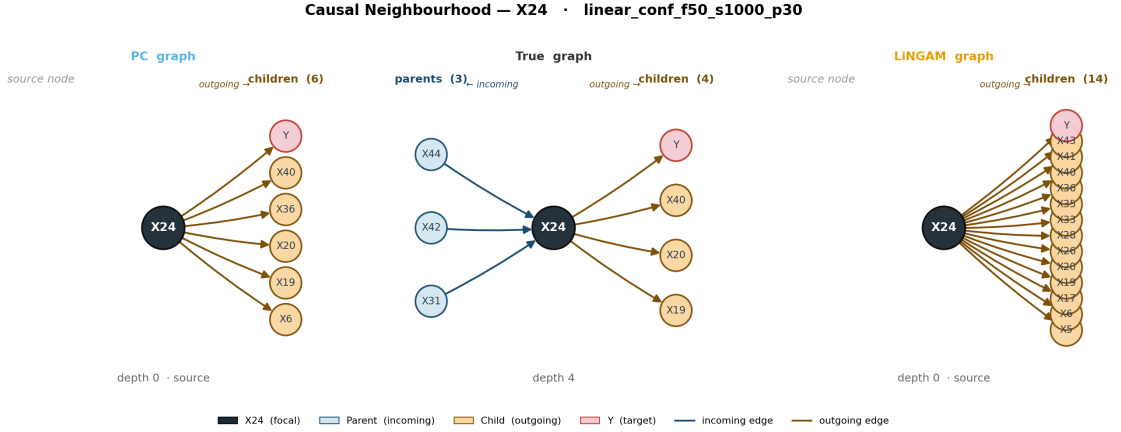


Figure 13: Causal neighbourhood of X24 across the PC (left), True DAG (centre), and LiNGAM (right) graphs, synthetic dataset. Blue edges are incoming (parents), amber edges are outgoing (children). In the True DAG, X24 is a direct parent of Y with 3 incoming and 4 outgoing edges. Both PC and LiNGAM strip its true incoming edges and add spurious outgoing ones (6 out-edges under PC, 14 under LiNGAM, 0 in-edges in both), recasting a mid-graph node as an apparent root source.

X24’s feature-level ΔM_{base} reaches 32.5% of $\hat{\sigma}$, the highest on the synthetic track, which is the first alarm signal flagging it for structural inspection. Both Flow+PC and Flow+LiNGAM produce inflated attributions relative to Flow+True: the instance-level signature of out-degree inflation, where spurious out-edges funnel additional credit through X24’s path-accounting. Notably, the second alarm signal does *not* fire: $\Delta M_{\text{disc}} \approx 8.9\%$ of $\hat{\sigma}$ is modest and X24 does not appear in the cross-discovery heatmap (Figure 8), because both PC and LiNGAM commit the same structural error and therefore agree with each other despite both deviating strongly from the True DAG. This exposes an important asymmetry in the diagnostic framework: ΔM_{base} fires whenever the injected graph pushes attributions away from the graph-free baseline, regardless of inter-algorithm agreement; ΔM_{disc} fires only when the two algorithms *disagree*. A high ΔM_{base} paired with a quiet ΔM_{disc} is therefore not a reassuring sign; it means the two algorithms have converged on the same wrong structure, a correlated

failure that the cross-discovery comparison cannot detect.

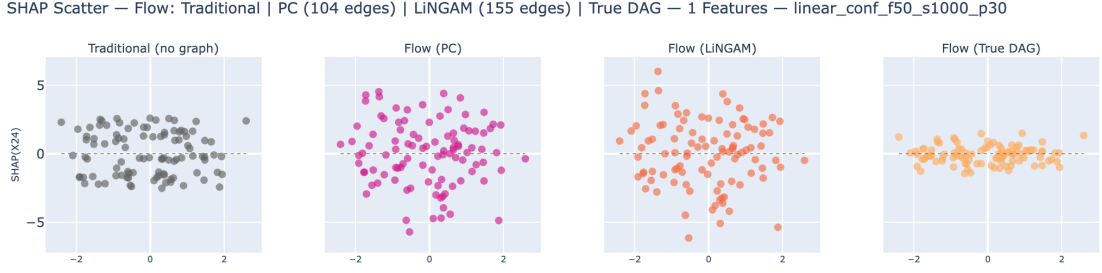


Figure 14: Shapley Flow SHAP scatter (feature value vs. SHAP value) for X24 under Traditional (no graph), Flow+PC, Flow+LiNGAM and Flow+True DAG, synthetic dataset. The Traditional column shows a stable attribution pattern; both discovered-graph columns diverge substantially from the True-DAG oracle, with inflated and partially sign-flipped values. The magnitude of this divergence, 32.5% of $\hat{\sigma}$, is precisely the ΔM_{base} signal identified in Section 4.2, making this scatter a direct visualisation of why large feature-level baseline divergence should trigger structural validation of the discovered graph.

X33 provides the contrasting case where both alarm signals fire simultaneously. Unlike X24, where PC and LiNGAM converge on the same error, X33’s neighbourhood is reconstructed differently by each algorithm: PC and LiNGAM disagree on both which parents X33 has and how many children it connects to, so neither graph reflects the true structure (Figure 15). This drives a high ΔM_{disc} , placing X33 among the top cross-discovery magnitude instability nodes on the synthetic track (Figure 8), while an equally high ΔM_{base} confirms that both discovered graphs inject substantial structural work on this node relative to the graph-free baseline. The Flow SHAP scatter (Figure 16) shows three distinct attribution clusters for Flow+PC, Flow+LiNGAM, and Flow+True DAG, none of which coincide: each graph assigns X33 a different role in the credit-routing mechanism and none recovers the oracle distribution. When both ΔM_{base} and ΔM_{disc} are elevated on the same node the diagnostic interpretation is unambiguous: the causal neighbourhood is both influential and contested, and a practitioner observing this two-signal pattern without a ground-truth graph has a clear directive to validate the specific edges flagged before trusting any structure-aware Shapley attribution for X33.

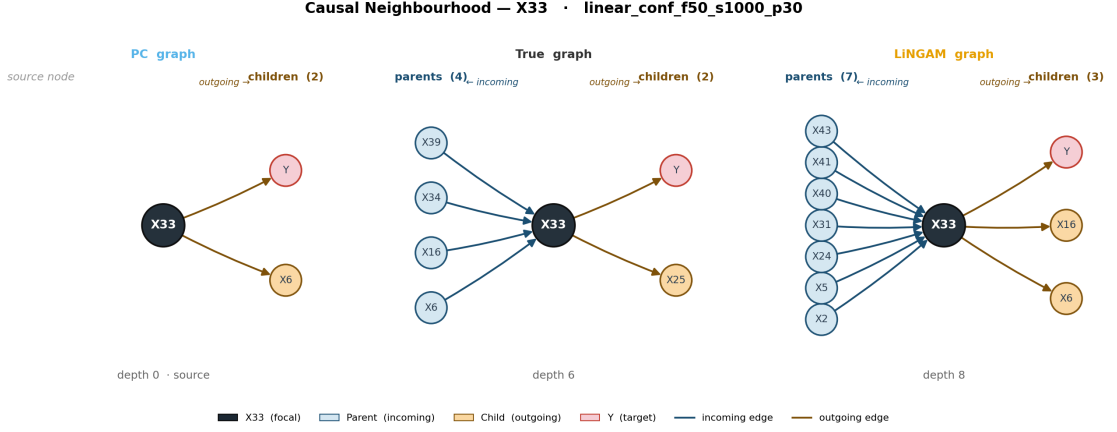


Figure 15: Causal neighbourhood of X33 across the PC (left), True DAG (centre), and LiNGAM (right) graphs, synthetic dataset. Neither PC nor LiNGAM recovers the true structure: the two algorithms disagree on both the parent set and the child set of X33, producing structurally divergent graphs that trigger both the ΔM_{base} and ΔM_{disc} alarm signals.

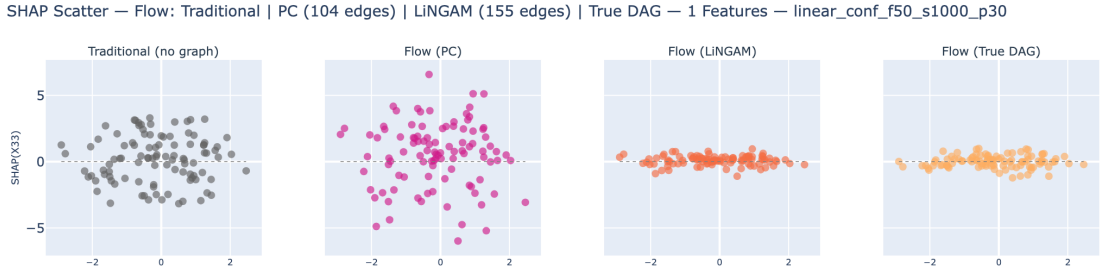


Figure 16: Shapley Flow SHAP scatter for X33 under Traditional (no graph), Flow+PC, Flow+LiNGAM, and Flow+True DAG, synthetic dataset. All three graph-aware columns produce distinct attribution distributions, none of which coincides with the oracle. This three-way divergence is the scatter-level signature of a node whose causal neighbourhood is both structurally influential and contested between the two discovery algorithms.

The same two-signal alarm pattern appears on the Sachs dataset through protein pka, a dominant near-root source (1 incoming, 6 outgoing edges in the consensus DAG). The first alarm fires at the feature level: ΔM_{base} reaches 17.3% of $\hat{\sigma}$ for Shapley Flow, placing pka among the highest-deviating proteins in the baseline heatmap (Figure 5). The second alarm fires at the cross-discovery level: ΔM_{disc} is also large, because PC and LiNGAM disagree substantially on pka’s neighbourhood (Figure 8). PC reverses several true outgoing edges and gives pka 3 incoming and 3 outgoing connections, while LiNGAM keeps it a near-root with 7 outgoing edges and none incoming (Figure 17). Both alarms active on the same node constitute a clear directive: the discovered causal connections of pka are both influential and contested, and the specific edges flagged should be validated (for instance by checking whether pka genuinely acts as a root

source or receives upstream inputs) before trusting any structure-aware attribution for this protein. The oracle confirms the damage: Flow+LiNGAM tracks the consensus DAG closely while Flow+PC inflates pka’s attribution substantially, with the scatter showing two distinct clusters that separate along the PC/LiNGAM axis (Figure 18).

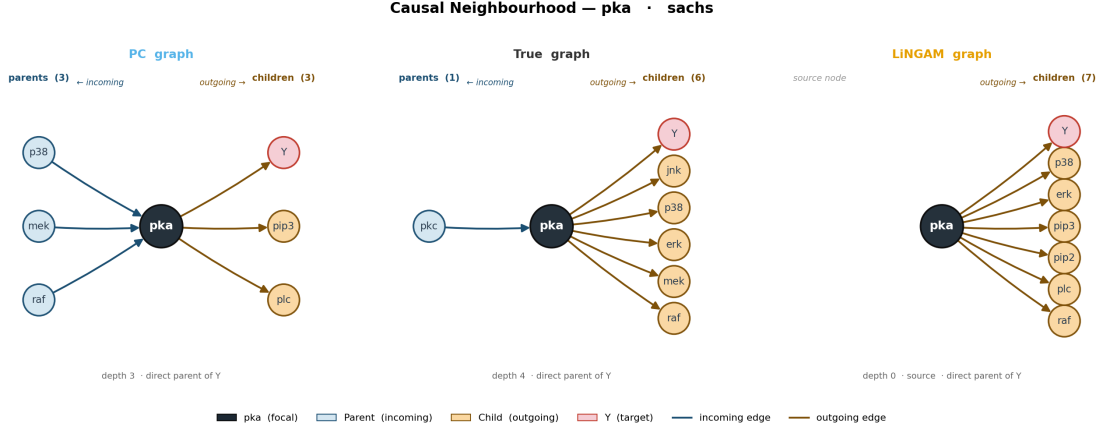


Figure 17: Causal neighbourhood of pka across the PC (left), True DAG (centre), and LiNGAM (right) graphs, Sachs dataset. pka is a near-root source with 1 incoming and 6 outgoing edges in the consensus DAG. PC gives pka 3 incoming and 3 outgoing edges (reversing several true outgoing links), while LiNGAM keeps it a pure root with 7 outgoing edges and none incoming, much closer to the consensus structure. The PC reversal that adds spurious incoming edges drives its ΔM_{oracle} from 6.4% of $\hat{\sigma}$ (LiNGAM) up to 25.0% (PC).

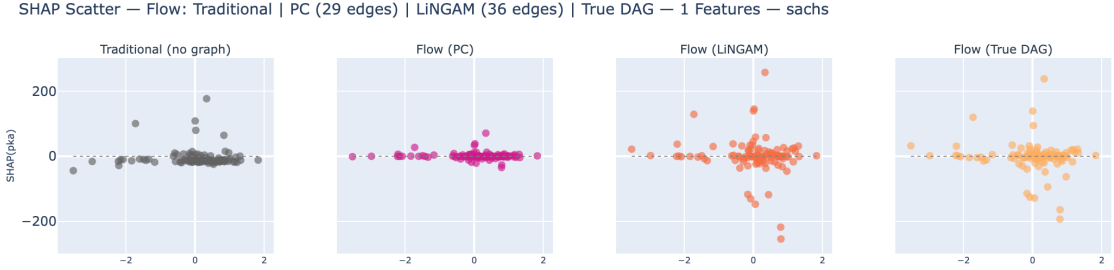


Figure 18: Shapley Flow SHAP scatter for pka, Sachs dataset. Flow+LiNGAM tracks Flow+True closely, while Flow+PC produces a visibly inflated attribution cluster, the scatter-level signature of PC’s edge-reversal error that recasts pka as a high-in-degree intermediate node rather than the near-root source it is in the consensus DAG.

4.5.2 Directed Edge Inversion and Causal Credit Transfer

This mechanism illustrates how a single edge orientation error by one discovery algorithm triggers the two-signal alarm on the affected node and produces a divergent attribution pattern relative to both the other algorithm and the oracle. The featured node is X47, a dominant direct parent of Y with 3 incoming and 1 outgoing edge in the True DAG.

Feature-level ΔM_{base} flags X47 as having elevated divergence from the Traditional Shapley baseline, and feature-level ΔM_{disc} places it among the nodes with the largest magnitude gap between the PC and LiNGAM variants (reaching 29.8% of $\hat{\sigma}$ for CSV). Both alarms fire simultaneously, the same two-signal pattern seen for X33, identifying X47 as a node where the discovered causal structure is both influential and contested.

Inspecting X47’s causal neighbourhood (Figure 19) reveals the cause. In the True DAG and under PC, X47 is a well-connected mid-graph node with 3 incoming edges. LiNGAM reverses those edges entirely, leaving X47 with 0 incoming and 5 outgoing edges and recasting it as an apparent root source. The reversal is not an isolated local error; it propagates through the edge $X_{21} \rightarrow X_{47}$ (correctly oriented by PC, reversed to $X_{47} \rightarrow X_{21}$ by LiNGAM), which transforms X47 from X21’s child into its apparent parent (Figure 20).

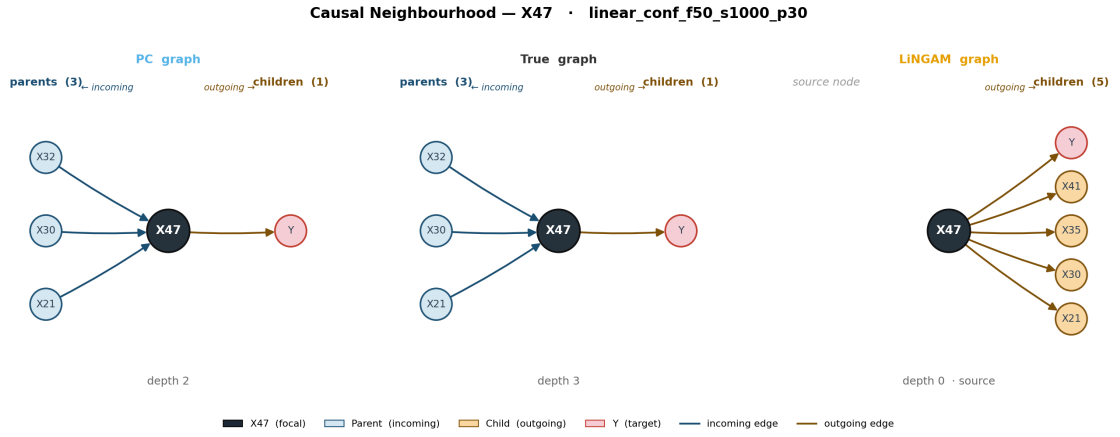


Figure 19: Causal neighbourhood of X47 across the PC (left), True DAG (centre), and LiNGAM (right) graphs, synthetic dataset. In the True DAG, X47 is a mid-graph node with 3 incoming and 1 outgoing edge. PC recovers its incoming edges accurately; LiNGAM reverses them, leaving X47 as a 0-in/5-out apparent root source. This is the structural error that triggers both the ΔM_{base} and ΔM_{disc} alarms.

The attribution consequences split sharply. Under PC, both Causal Shapley and Shapley Flow correctly treat X47 as a downstream node: it receives credit proportional to its true direct contribution to Y , and X21 receives indirect credit for its upstream influence through X47. Under LiNGAM the reversed edge places X47 upstream of X21; in Causal Shapley this injects X47 into X21’s interventional parent set, contaminating both attributions, while in Flow the outgoing edge $X_{47} \rightarrow X_{21}$ routes X47’s credit toward X21, reducing X47’s net attribution and inflating X21’s. The result is that X47 is under-attributed under LiNGAM relative to the oracle, while PC and True DAG agree closely.

The practical lesson is that the appropriate response to this two-signal pattern is not to

prune the edge between X_{21} and X_{47} , the edge is genuine, but to invest in resolving its correct orientation before running the structure-aware Shapley computation. Removing a real causal link would suppress legitimate indirect-effect attribution; correcting the orientation (as PC achieves here) restores oracle-consistent explanations without discarding structural information.

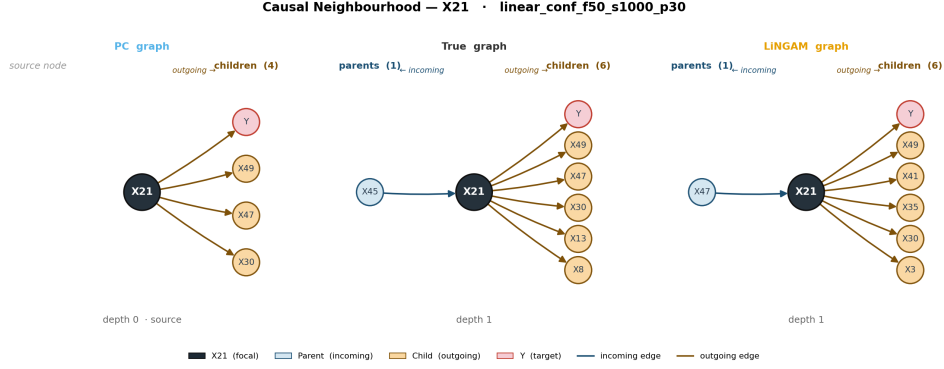
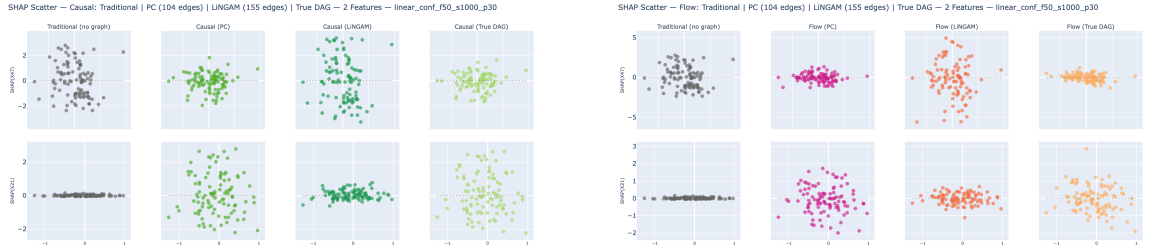


Figure 20: Causal neighbourhood of X_{21} across the PC (left), True DAG (centre), and LiNGAM (right) graphs, synthetic dataset. In the True DAG and under PC, the edge runs $X_{21} \rightarrow X_{47}$. LiNGAM reverses it to $X_{47} \rightarrow X_{21}$, placing X_{47} upstream of X_{21} and misdirecting attribution credit in both Causal Shapley and Flow.



(a) Causal Shapley: under PC (correct orientation) both X_{21} and X_{47} attributions track the oracle. Under LiNGAM (reversed orientation), X_{47} is suppressed and X_{21} acquires inflated credit.

(b) Shapley Flow: the same orientation-driven credit transfer appears; Flow+PC and Flow+True agree closely, while Flow+LiNGAM diverges on both features.

Figure 21: SHAP scatter for X_{21} and X_{47} across graph configurations.

4.5.3 Structural Role Error Resolved by Domain Knowledge

Protein erk on Sachs is the node in this analysis where both observable alarm signals fire simultaneously and with the highest intensity of the entire experiment. The first alarm fires at the feature level: ΔM_{base} for erk is elevated across all three Shapley methods in the baseline heatmap (Figure 5), indicating that both discovered graphs are doing substantial structural work on erk’s attribution relative to the graph-free baseline. The second alarm fires at the cross-discovery level: ΔM_{disc} for erk reaches 57.5% of $\hat{\sigma}$ under Shapley Flow and 35.8% even under Asymmetric Shapley (Figure 8),

the latter being the highest ΔM_{disc} that ASV produces for any feature on either dataset. A practitioner running both algorithms and comparing feature-level heatmaps would flag erk immediately: both alarms active on the same protein constitute a clear directive to inspect its discovered neighbourhood before trusting any structure-aware attribution. Inspecting the neighbourhood (Figure 22) reveals what drives the disagreement. PC removes erk’s incoming edges entirely, isolating it as an apparent root node with outgoing edges only. LiNGAM, by contrast, recovers at least some incoming connections for erk, placing it closer to the downstream position it occupies in the consensus DAG. The two discovered graphs therefore disagree fundamentally on erk’s role: source versus intermediate node, and that structural disagreement is precisely what ΔM_{disc} measures.

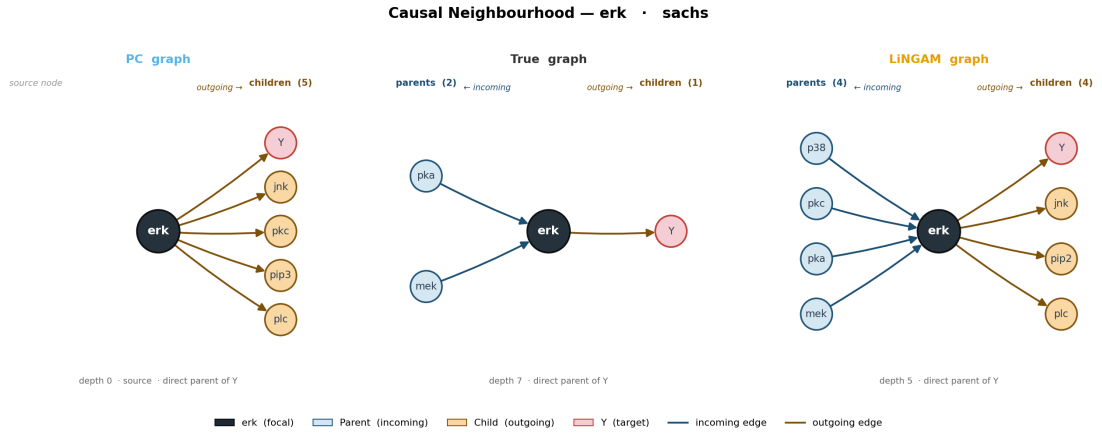


Figure 22: Causal neighbourhood of erk across the PC (left), True DAG (centre), and LiNGAM (right) graphs, Sachs dataset. In the consensus DAG erk is a downstream node receiving inputs from pka and mek. PC removes its incoming edges and leaves it as a boundary source; LiNGAM recovers at least some incoming connections, placing it closer to its true downstream role. This structural divergence between the two algorithms is what triggers both the ΔM_{base} and ΔM_{disc} alarms.

With the two alarms flagging erk, a practitioner has a concrete target for validation. Consulting the biological literature or a domain expert would confirm that erk is indeed a downstream signaling protein receiving input from pka and mek, not a root source. This external knowledge, combined with erk’s known direct causal role as a predictor of Akt, disambiguates the two discovered graphs: LiNGAM’s partial recovery of incoming edges for erk is structurally closer to reality than PC’s full isolation. The ΔM_{oracle} metric confirms this: under Shapley Flow, Flow+PC produces a feature-level oracle deviation of 49.2% of $\hat{\sigma}$, the largest single oracle deviation in the entire experiment, while Flow+LiNGAM is substantially closer to the consensus DAG reference. The SHAP scatter (Figure 23) makes the instance-level consequence visible: Flow+PC assigns erk a consistently inflated attribution cluster consistent with source-node overloading, while Flow+LiNGAM produces a less extreme distortion that moves in the right direction

relative to the oracle. The practical takeaway matches the other two-signal cases: the alarms identified erk as a node requiring structural validation without any oracle access, and the subsequent validation confirmed which discovered algorithm should be trusted for this protein.

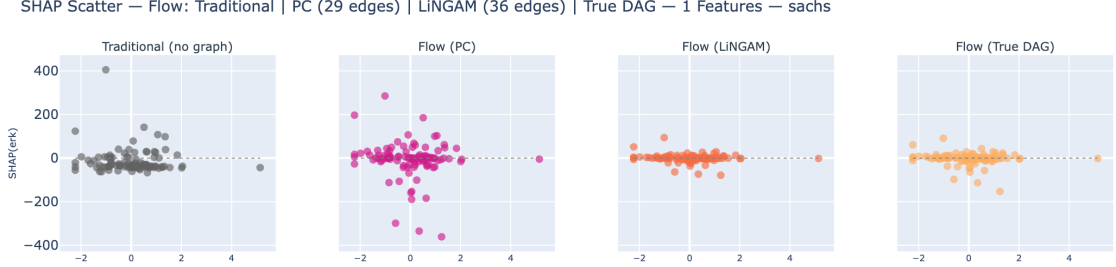


Figure 23: Shapley Flow SHAP scatter for erk, Sachs dataset. Flow+PC produces a heavily inflated attribution cluster consistent with PC treating erk as an isolated root source. Flow+LiNGAM is less distorted and sits closer to the oracle distribution, reflecting LiNGAM’s partial recovery of erk’s incoming edges. The gap between the two discovered-graph columns is the scatter-level signature of the ΔM_{disc} alarm, and the oracle column confirms that LiNGAM’s graph is the more trustworthy input for this protein.

4.5.4 Near-Zero Attributions and Sign Disagreement Explosion

The magnitude cases above focus on nodes whose absolute attributions diverges substantially when a causal graph is injected. The sign heatmaps tell a different story: the features with the highest D_{base} and D_{disc} are a completely separate set from the magnitude leaders, and the reason is structural. A feature whose Traditional Shapley value already sits close to zero needs only a small graph-induced perturbation to cross the sign boundary; the same perturbation applied to a high-magnitude feature would leave its sign untouched. This section illustrates the mechanism with two concrete instances.

On the synthetic dataset, X10 is the clearest example. Its Traditional Shapley attributions sit near zero across the test set, so PC’s structural changes to X10’s neighbourhood (Figure 24) are enough to push most instances across the sign boundary, producing a feature-level D_{base} of 96% for Causal+PC, the highest in the synthetic experiment. The SHAP scatter (Figure 25) makes this visible: a tight near-zero Traditional cluster fans out into a broad, sign-mixed Causal+PC distribution. The D_{disc} for X10 would show a high sign disagreement between PC and LiNGAM, confirming that the discovered neighbourhood is contested and that any structure-aware attribution for this feature should be treated with caution until the graph is validated.

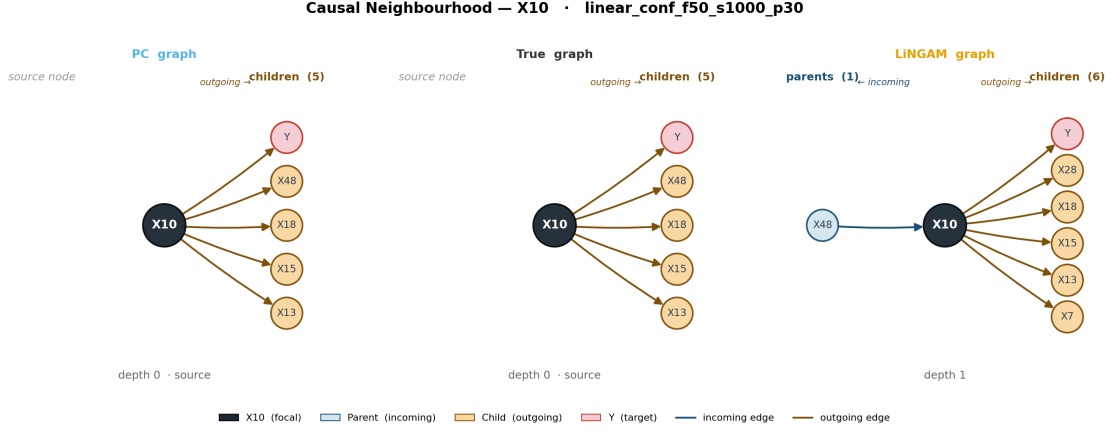


Figure 24: Causal neighbourhood of X10 across the PC (left), True DAG (centre), and LiNGAM (right) graphs, synthetic dataset. X10’s neighbourhood is reconstructed differently by PC relative to the True DAG, introducing structural changes that shift its interventional conditioning set. Because X10’s Traditional Shapley values sit near zero, even small attribution shifts are sufficient to flip sign on nearly all instances.

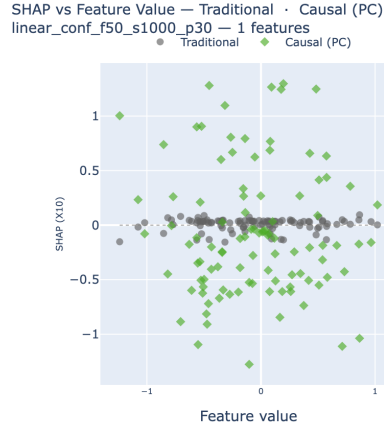


Figure 25: Traditional vs. Causal+PC SHAP scatter for X10, synthetic dataset. The Traditional column shows attributions clustered near zero; the Causal+PC column disperses them to both sides of zero with larger magnitude. This scatter-level divergence, producing $D_{\text{base}} = 96\%$, is the direct consequence of near-zero baseline attributions being pushed across the sign boundary by graph-induced interventional redistribution.

The same mechanism appears on Sachs through protein *pkc*, which leads the sign heatmaps (D_{base} and D_{disc} highest among Sachs proteins under Causal Shapley) yet is invisible in the magnitude heatmaps. Its neighbourhood (Figure 26) is disputed between PC and LiNGAM, but because its Traditional values sit near zero the structural changes need only be small to flip sign on a large fraction of instances, as the SHAP scatter (Figure 27) confirms.

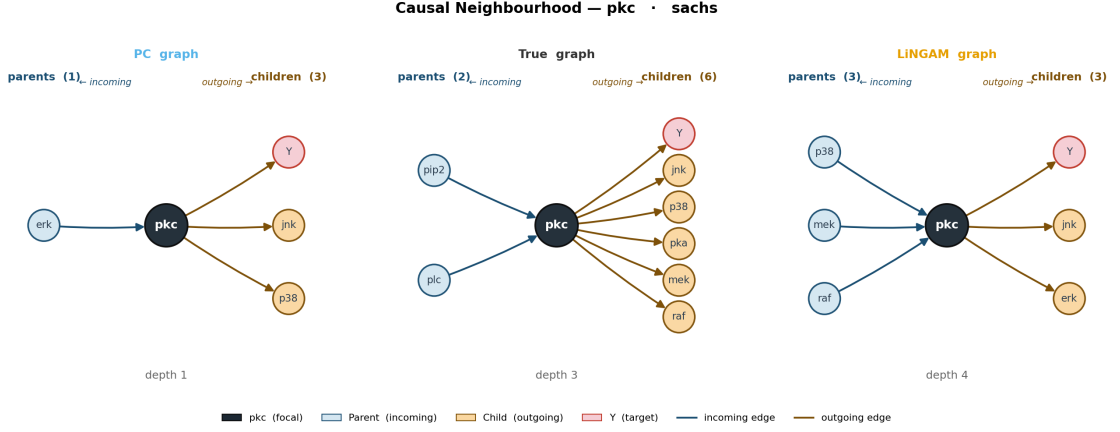


Figure 26: Causal neighbourhood of pkc across the PC (left), True DAG (centre), and LiNGAM (right) graphs, Sachs dataset. PC and LiNGAM discover different parent and child sets for pkc, generating different interventional conditioning environments. Because pkc’s Traditional Shapley values sit near zero, these structural differences are sufficient to produce high sign disagreement without producing large magnitude divergence.

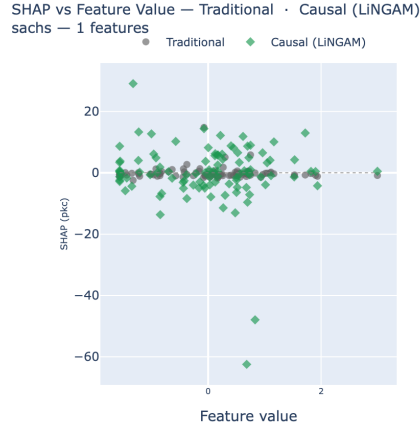


Figure 27: Traditional vs. Causal+LiNGAM SHAP scatter for pkc, Sachs dataset. The Traditional column concentrates near zero; Causal+LiNGAM disperses attributions to both sides of zero. The magnitude of the shift is small relative to $\hat{\sigma}$, which is why pkc does not appear in the magnitude heatmaps, but it is large enough to cross the sign boundary for a high fraction of instances, which is precisely why it leads the sign heatmaps instead.

Together X10 and pkc show that $(D_{\text{base}}, D_{\text{disc}})$ and $(\Delta M_{\text{base}}, \Delta M_{\text{disc}})$ are complementary diagnostics: the magnitude metrics flag nodes whose attribution size changes, while the sign metrics surface a different set of nodes, ones that were near-invisible under Traditional Shapley but now acquire meaningful (and potentially inverted) explanatory weight once a graph is introduced. For a practitioner this matters: these features were not on the radar before, and the sign heatmap is the only signal that brings them to attention.

Taken together, the four case studies validate the two-signal alarm workflow described at the opening of this section. In each instance the practitioner-visible signals, ΔM_{base} , ΔM_{disc} , D_{base} , and D_{disc} , identified the structurally fragile nodes without oracle access; the oracle comparison confirmed afterwards that the signals were pointing at genuine problems. The cases also illustrate that no single alarm pattern is sufficient on its own: a high ΔM_{base} paired with a quiet ΔM_{disc} reveals a correlated failure where both algorithms converge on the same wrong structure (X24); simultaneous elevation on both magnitude alarms marks a fully contested neighbourhood (X33, pka, erk); and high D_{base} or D_{disc} without a matching magnitude spike unmasks a distinct population of near-zero features that would otherwise escape attention entirely (X10, pkc). Running all four metrics together is what makes the framework diagnostic rather than merely descriptive.

5 Conclusion

5.1 Summary of Findings

This thesis built an experimental pipeline to give practitioners a structured approach for cases where their modelled system has causal relationships and they want structure-aware Shapley explanations that are more faithful to those dependencies.

The pipeline tests three structure-aware Shapley methods, Asymmetric Shapley (ASV), Causal Shapley (CSV), and Shapley Flow, under two automatically discovered graphs (PC and LiNGAM) across three comparison tracks: against the graph-free Traditional Shapley baseline, as cross-discovery sensitivity between PC and LiNGAM, and against the True DAG oracle. The evaluation framework introduces two complementary dimensions: magnitude of change (ΔM) and sign change (D), each measured at three levels, global (dataset), feature, and instance, forming a systematic diagnostic alert system whose behaviour is illustrated through the most informative high-deviation cases.

The introduced experimental metrics, were tested in two datasets, a synthetic linear confounded dataset with 50 features and a sparse causal structure, and the real-world Sachs dataset with 10 proteins and a denser causal structure. The results show that the introduced metrics successfully flag structurally fragile nodes without oracle access, guiding practitioners on where to focus validation efforts when working with imperfect causal graphs.

The three methods occupy distinct sensitivity regimes, as expected, but the experimental pipeline now makes those differences measurable. ASV is stable under graph errors: on synthetic, $\Delta M_{\text{base}} < 1\%$ of $\hat{\sigma}$ and $D_{\text{base}} < 6\%$, and cross-discovery sensitivity stays

below 1.1% and 6% respectively. On Sachs the numbers rise, but the framework remains informative: feature-level ΔM_{disc} and D_{disc} point directly to erk and pka as the contested nodes. CSV and Flow are more exposed because they embed the graph deeper into the computation, CSV inside the value function, Flow in the players themselves, so their baseline and cross-discovery divergences are substantially higher than ASV’s on both datasets.

Two patterns hold across all settings. First, the specific discovered graph supplied (PC or LiNGAM) has little effect on the baseline deviation: both perturb ΔM_{base} and D_{base} by nearly identical amounts for a given method, so what matters is whether a causal graph is used at all, not which specific graph is supplied. Second, sign disagreement (D) is a more severe perturbation than magnitude divergence (ΔM): adding a causal graph flips attribution directions more often than it inflates their size, because near-zero attributions need only a small shift to cross the sign boundary whereas large attributions absorb the same redistribution without changing direction.

In real-world scenarios the experimental pipeline does not reveal which graph structure is truly correct: the discovered graph will always be imperfect. When the causal structure of the observational data is unclear, the safe choice is ASV, which absorbs discovery errors rather than inheriting them. CSV and Flow should only be used when the cross-discovery metrics (ΔM_{disc} , D_{disc}) are low across all features, indicating that any reasonable graph produces similar attributions and no contested nodes exist.

The metrics surface three distinct alarm patterns visible in the case studies. ΔM_{base} alone flags a shared graph failure: X24 reaches $\Delta M_{\text{base}} = 32.5\%$ of $\hat{\sigma}$ while $\Delta M_{\text{disc}} \approx 8.9\%$ stays quiet, because both discovery algorithms commit the same structural error and agree with each other despite both being wrong, this would be a blind spot for practitioners. ΔM_{base} and ΔM_{disc} firing together mark a contested neighbourhood: pka ($\Delta M_{\text{base}} = 17.3\%$ for CSV PC, $\Delta M_{\text{disc}} = 30.2\%$ for ASV), X47 ($\Delta M_{\text{base}} = 21.8\%$ for CSV PC, $\Delta M_{\text{disc}} = 29.7\%$ for CSV), and erk ($\Delta M_{\text{base}} = 57.9\%$ for Flow PC, $\Delta M_{\text{disc}} = 57.5\%$ for Flow and even $\Delta M_{\text{disc}} = 35.8\%$ for ASV) all show both alarms simultaneously, identifying nodes where the discovered causal structure is influential and disputed. The sign metrics surface a third class: X10 reaches $D_{\text{base}} = 96\%$ for Causal+PC and pkc leads the Sachs sign heatmaps, yet neither appears in the magnitude heatmaps, both carry near-zero baseline attributions, so a small structural shift is enough to flip their direction without leaving a large RMS footprint. Running all four metrics together is therefore what makes the framework diagnostic rather than merely descriptive: each alarm pattern reveals a structurally distinct failure mode that the others cannot detect.

5.2 Main Contribution and Practical Recommendations

The most important contribution of this work is the evaluation framework itself. Before this work, a practitioner who wanted to add a discovered causal graph to a Shapley computation had no systematic way to assess what that decision would cost in attribution quality. The metrics defined here, ΔM_{base} , ΔM_{disc} , D_{base} , D_{disc} , at both global level and feature level, give a practitioner a structured set of tools for the assessment. Each metric measures a different dimension of change: how much the graph changes attributions away from the graph-free model behaviour, and how stable those attributions are when the input causal graph is swapped. Together these two dimensions describe not just whether attributions changed but whether the explanation can be trusted: a high baseline deviation signals that the graph is doing substantial structural work on a node, and a high cross-discovery instability signals which specific features would need additional revision of their discovered causal relationships before the attributions can be trusted. Both signals are computable without a ground-truth graph, ΔM_{base} and D_{base} only need the Traditional Shapley run as a reference, and ΔM_{disc} and D_{disc} only need two discovery algorithms run on the same data.

One practical use of the framework is to run two or more discovery algorithms on the same data and measure the cross-discovery stability (ΔM_{disc} , D_{disc}) before committing to any structure-aware Shapley method. If PC and LiNGAM produce very different attributions for the same feature, that is a direct sign that the method’s output depends more on which algorithm happened to be chosen than on the model’s actual predictive behaviour. The pipeline is therefore a diagnostic tool: it points to exactly which nodes and which edges are responsible for the instability, so graph refinement effort can be focused where it matters. Once the graph has been refined and a method chosen, reporting ΔM_{base} and D_{base} quantifies the net divergence of the final structure-aware explanation from the graph-free baseline, making the causal modelling decision transparent and auditable. The sign metrics extend this audit further: ΔM_{disc} flags nodes whose attribution magnitude is contested, D_{disc} identifies a distinct population of near-zero features whose explanatory direction flips between discovered graphs, features that would be entirely invisible in a magnitude-only inspection.

Given all of this, the practical ordering of the three methods is clear:

- **ASV is the recommended default.** It delivers oracle-level alignment with minimal deviation from the Traditional Shapley baseline, tolerates poor discovery results, and adds no computational cost above standard Monte Carlo SHAP. Even so, the diagnostic framework should always be run first: there can be contested features even under ASV that require graph validation before the attributions can be trusted.

- **CSV and Flow are useful only when graph quality is high** for example when domain experts have validated the structure or when the cross-discovery stability metrics confirm the explanations are graph stable.

5.3 Limitations and Future Work

The primary analysis was conducted on a single synthetic functional form (linear confounded) and one real dataset. Nonlinear and mixed functional forms, different graph densities, and larger sample sizes remain to be tested.

All Shapley computations use $T = 100$ Monte Carlo samples, a deliberate trade-off for the 50-feature system and topological order loops for the Causal Shapley. Increasing T would reduce sampling noise. The CSV implementation uses a conditional Gaussian approximation ($M = 10$ inner samples), which is appropriate for linear systems but likely underperforms on nonlinear data.

Future work should extend the evaluation to nonlinear Shapley variants, explore ensemble approaches that average attributions across multiple discovered graphs to reduce cross-discovery instability, and investigate whether active learning strategies could use the feature-level ΔM_{disc} signal to guide targeted graph refinement, starting from the nodes the framework identifies as most contested.

Declaration on the Use of Artificial Intelligence

All research questions, experimental designs, analyses, tables, figures, and conclusions presented in this thesis are the original work of the author. Artificial intelligence tools were used exclusively as writing aids to check grammar, improve textual cohesion, and refine word choice. No AI tool was used to generate, interpret, or derive any of the scientific content, data, or ideas contained herein.

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Appendix: Shapley Method Subroutines

This appendix collects the helper subroutines invoked by the algorithm blocks of Section 3. Each subroutine is grouped under the method or methods that use it. The main algorithm blocks in Section 3 call these by name.

A.1 Coalition Value (Traditional and Asymmetric Shapley)

Both Traditional Shapley (Section 3) and Asymmetric Shapley estimate the observational coalition value $v(S) = \mathbb{E}[f(X) \mid X_S = x_S]$ with the same subroutine. Features in the coalition S take the instance’s real values; the remaining features are filled in from each background row, and the model output is averaged over the background.

Algorithm 5 COALITION_VALUE — $v(S) = \mathbb{E}[f(X) \mid X_S = x_S]$

```

1: function COALITION_VALUE( $x, S, f, D$ )
2:   samples  $\leftarrow$  copy( $D$ ) ▷ one row per background instance
3:   for each feature  $j \in S$  do
4:     samples[:,  $j$ ]  $\leftarrow$   $x[j]$  ▷ overwrite column  $j$  with the real value
5:   end for return mean( $f(\text{samples})$ ) ▷ average prediction over the background
6: end function

```

A.2 Uniform Random Topological Ordering (Asymmetric and Causal Shapley)

Asymmetric Shapley and the outer loop of Causal Shapley both draw a uniform random linear extension of a DAG with a randomized Kahn algorithm. It maintains a pool of “ready” nodes whose ancestors have all been placed and selects one uniformly at each step, which samples uniformly over the linear extensions and runs in $O(n)$ per ordering. The children lists and initial in-degrees are precomputed once over the relevant DAG (the X -only feature subgraph for Asymmetric; the component DAG for Causal; Y is excluded, as it is never a player). For Asymmetric the procedure operates on features; for Causal it operates on components, whose ordering is then expanded to features.

Algorithm 6 SAMPLE_TOPOLOGICAL_ORDERING — uniform random linear extension (Kahn)

PRECOMPUTE: (once): `children[], in_degree[]` from the DAG edges (Y excluded)

If the feature DAG contains a cycle, disable constraints $\Rightarrow$ reduces to Traditional Shapley

```

1: function SAMPLE_TOPOLOGICAL_ORDERING(children, in_degree)
2:   deg  $\leftarrow$  copy(in_degree)
3:   ready  $\leftarrow$  [i : deg[i] = 0] ▷ source nodes
4:   order  $\leftarrow$  []
5:   while ready  $\neq \emptyset$  do
6:     k  $\leftarrow$  UNIFORM_RANDOM_INDEX(ready)
7:     node  $\leftarrow$  ready[k]; swap-remove from ready ▷  $O(1)$  swap-remove
8:     order.append(node)
9:     for child  $\in$  children[node] do
10:      deg[child]  $\leftarrow$  deg[child] - 1
11:      if deg[child] = 0 then ready.append(child)
12:      end if
13:    end for
14:  end while return order ▷ a uniform linear extension
15: end function

```

A.3 Post-Interventional Sampling (Causal Shapley)

The inner loop of Causal Shapley draws each sample from the post-interventional distribution $P(X \mid \text{do}(X_S = x_S))$ with the subroutine below. The intervened features are fixed to their instance values, and the remaining features are filled in component by component in a fixed topological order so that parents are resolved before children. Inside a confounded component the missing features are drawn independently given their parents, which destroys the spurious within-component correlation; in an ordinary component they are drawn jointly given the parents and any fixed siblings.

Algorithm 7 POST_INTERVENTIONAL_SAMPLE — draw from $P(X \mid \text{do}(X_S = x_S))$

```

1: function POST_INTERVENTIONAL_SAMPLE( $x, S, \text{components}, \text{confounded}[], \text{parents}[], \mu, \Sigma$ )
2:    $\text{sample} \leftarrow \mathbf{0}_n$ ; for  $j \in S$ :  $\text{sample}[j] \leftarrow x[j]$  ▷ fix the interventions
3:   for each component  $C$  in FIXED topological order do
4:      $\text{missing} \leftarrow C \setminus S$ 
5:     if  $\text{missing} = \emptyset$  then continue
6:     end if
7:     if  $\text{confounded}[C]$  then ▷ intervention breaks ties
8:       for  $j \in \text{missing}$  do ▷ draw INDEPENDENTLY
9:          $\text{sample}[j] \leftarrow \text{GAUSSIAN\_CONDITIONAL}(\text{target} = \{j\}, \text{cond} = \text{parents}[C])$ 
10:      end for
11:     else ▷ ordinary dependence
12:        $\text{fixed} \leftarrow C \cap S$  ▷ draw JOINTLY given parents AND siblings
13:        $\text{sample}[\text{missing}] \leftarrow \text{GAUSSIAN\_CONDITIONAL}(\text{target} = \text{missing}, \text{cond} = \text{parents}[C] \cup \text{fixed})$ 
14:     end if
15:   end for return sample
16: end function

```

The conditional draws use the closed-form Gaussian expression under a multivariate-Gaussian approximation of the background data, $X \sim \mathcal{N}(\mu, \Sigma)$. A singular conditioning matrix falls back to the conditional mean, and the univariate branch is used whenever a single feature is sampled (the only case exercised in this thesis, since the confounder list is empty).

Algorithm 8 GAUSSIAN_CONDITIONAL — closed-form conditional Gaussian draw

```

1: function GAUSSIAN_CONDITIONAL( $\text{target } A, \text{cond } B, \text{vals } b$ )
2:   if  $B = \emptyset$  then return draw from  $\mathcal{N}(\mu_A, \Sigma_{AA})$  ▷ marginal
3:   end if
4:    $\Sigma_{BB}^- \leftarrow \text{PSEUDO\_INVERSE}(\Sigma_{BB})$ 
5:    $\mu_{\text{cond}} \leftarrow \mu_A + \Sigma_{AB} \Sigma_{BB}^- (b - \mu_B)$ 
6:    $\Sigma_{\text{cond}} \leftarrow \Sigma_{AA} - \Sigma_{AB} \Sigma_{BB}^- \Sigma_{BA}$ 
7:   symmetrise  $\Sigma_{\text{cond}}$ ; nudge eigenvalues  $\geq \varepsilon$  return draw from  $\mathcal{N}(\mu_{\text{cond}}, \Sigma_{\text{cond}})$  ▷ scalar branch if  $|A| = 1$ 
8: end function

```

A.4 System Value (Shapley Flow)

Shapley Flow evaluates the model on a set of active edges with the subroutine below. Each node is assigned its foreground or background value according to the activation rule, Y is stripped before the model is evaluated, and the empty edge set returns $f(x_{\text{bg}})$ while the full edge set returns $f(x_{\text{fg}})$.

Algorithm 9 SYSTEM_VALUE — binary foreground/background activation

```

1: function SYSTEM_VALUE(active_edges,  $x_{\text{fg}}$ ,  $x_{\text{bg}}$ )
2:   for each node  $i$  in the graph do
3:     if  $i$  is a source and has an active outgoing edge then  $\text{val}[i] \leftarrow x_{\text{fg}}[i]$ 
4:     else if  $i$  has an active incoming edge then  $\text{val}[i] \leftarrow x_{\text{fg}}[i]$ 
5:     else if edge  $(i \rightarrow Y)$  is active then  $\text{val}[i] \leftarrow x_{\text{fg}}[i]$ 
6:     else  $\text{val}[i] \leftarrow x_{\text{bg}}[i]$ 
7:     end if
8:   end for
9:   drop  $Y$  from  $\text{val}$   $\triangleright Y$  is never an input to  $f$  return  $f(\text{val})$ 
10: end function

```
